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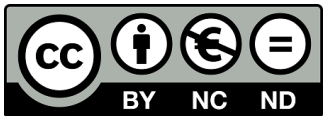
Design and implementation of an AI-powered English-Spanish Idiom Learning System

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Leganés, September 2026



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SUMMARY

Idiomatic expressions are one of the areas of language learning that many people may not pay sufficient attention to. This challenging aspect of mastering a foreign language was chosen as the main motivation for this project. The project highlighted the importance of idiom learning through the implementation of an AI-powered English–Spanish Idiom Learning System designed to support language learning through personalized and interactive learning activities.

The application was developed using Python, initially deployed using Google Colab, VS Code, and finally on Hugging Face Spaces, with Streamlit as its main framework. Several Natural Language Processing (NLP) technologies were integrated, such as MarianMT for literal translation, FLAN-T5-base for AI-generated quizzes, Multilingual MiniLM for semantic idiom detection, and Google Text-to-Speech (gTTS) for complementary audio pronunciation. In addition, SQLite was deployed in the application as the main storage system, supporting user learning statistics and adaptive quizzes.

The application offered interactive learning through five different modes, including idiom exploration with bilingual idiom explanations, free-text idiom detection, quizzes, favorites management, and a learning analytics system.

The outcome of the system is an idiom-learning application supported by personalized practice, bilingual translations, and artificial intelligence. The project thus contributes by concentrating on colloquial idioms and cross-linguistic equivalences between English and Spanish, offering a specialized alternative to general language-learning tools.

Keywords: Artificial Intelligence, Natural Language Processing, FLAN-T5, MarianMT, MiniLM, Idiom Learning, Streamlit, SQLite, Hugging Face, Idiom Learning.

DEDICATION

After months of working on this project, day and night, it has finally come to an end. There have been a lot of challenges along the way to make this project a reality. It might look simple to others, but only I know how many failures I encountered throughout the journey. At one point, I even wanted to change everything and start all over again.

I want to thank my advisor, Pedro Manuel Moreno Marcos, for all the support and help in making my work come to realization.

To all my professors at Universidad Carlos III de Madrid, thank you for the knowledge they have imparted to me throughout my entire academic career and for the trust they gave me to never stop trying.

To all my friends at the university, as we convinced each other from the first year to have faith and confidence that this is the right path for us. We have each other; we are survivors.

To my family, who always give me motivation, as I am the only one in my family pursuing a STEM field, completely different from theirs and known to be very challenging.

To my best friend, Najla Nabila, who kindly listens to all my experiences and rants throughout the journey of studying abroad. Somehow, as time went by, it just became harder to bear. But I managed it, Alhamdulillah.

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1. INTRODUCTION

1.1. Motivation of the project

Second-language learners face several specific obstacles in interpreting idioms. Although learners may recognize the figurative meaning of an idiom, they often revert to a literal interpretation [1]. A limited vocabulary prevents them from employing the same word-stretching techniques that native speakers utilize to interpret figurative meanings [2]. Moreover, many idioms originate from particular cultural or historical backgrounds that cannot be understood solely by examining their individual words [3].

Traditional language-learning materials, such as dictionaries and textbooks, often offer explanations and illustrations of idiomatic expressions; however, they frequently do not include engaging and interactive elements that promote active learning and personalized practice. Consequently, learners might find it challenging to identify idiomatic expressions in authentic contexts and use them correctly in conversations.

Recent developments in Natural Language Processing (NLP) and Artificial Intelligence (AI) have created new opportunities for the development of intelligent language-learning applications [4]. Pre-trained language models and semantic similarity techniques enable tasks such as idiom detection [5], translation with integrated audio pronunciation, and adaptive quiz generation, offering learners a more engaging learning experience.

Motivated by these challenges and opportunities, as well as recent advancements in NLP, this project aims to develop an AI-powered English–Spanish idiom-learning application that combines educational resources with NLP technologies. The application provides an interactive learning environment within a single platform, including idiom exploration, semantic idiom detection, audio pronunciation, adaptive quizzes, and learning analytics.

1.2. Objectives

The main objective of this project is to design and develop an AI-powered English–Spanish idiom-learning application. This application helps second-language learners of certain level understand the figurative and cultural context of idioms in everyday conversations by differentiating between English and Spanish idioms that may share similar meanings or contexts.

English is the most widely spoken language in the world when both native and non-native speakers are considered, with approximately 1.5 billion speakers [6]. Meanwhile, Spanish is the fourth most spoken language worldwide and the second most spoken native language, surpassing English in terms of native speakers, with over 485 million native

speakers [7].

To achieve this general objective, the following specific objectives (O1–O5) were defined:

O1. Create a multilingual idiom dataset, structured with a focus on the differentiation and usage of both languages based on real-world examples.

O2. Develop mode that helps detect idioms in free-form sentences written by the user, either in English or Spanish.

O3. Design and implement a feature that helps users better differentiate idiom meanings by introducing literal translations alongside their natural idiomatic meanings, making the gap between figurative and literal language more explicit.

O4. Develop an AI-powered quiz generation mode featuring example questions with multiple idioms to choose from, together with adaptive questions based on the user's performance.

O5. Deploy a fully functional web application with several features to support idiom learning, such as idiom exploration, idioms in sentences, adaptive quizzes, favorites, and learning analytics features.

1.3. Structure of the report

The following list describes the structure of this report to facilitate comprehension.

- **Chapter 1: Introduction**

  Presents the motivation, objectives, and overall structure of the project.

- **Chapter 2: Theoretical Framework**

  Overview of idioms and language learning, neural machine translation, large language models, development technologies, and other tools.

- **Chapter 3: Methodology**

  Describes the datasets, data processing procedures, system architecture, and AI models used in the project.

- **Chapter 4: System Implementation**

  Presents the application architecture and application modes, including Explore Idioms, Idioms in Sentences, Quiz, Favorites, and Learning Analytics, as well as the graphical interface.

- **Chapter 5: Results and Evaluation**

  Describes functional testing, AI model evaluation, and user experience evaluation.

- **Chapter 6: Conclusions**

  Summarizes the application results, highlights the project's achievements, discusses its limitations, and proposes future improvements.

2. THEORETICAL FRAMEWORK

2.1. Idioms and Language Learning

Definition of idioms

Idioms, or in Spanish "los modismos", are an important component of the language. The term originates from the Greek word "idiōma", which in the fifteenth century meant a "form of speech peculiar to a people or place". In the sixteenth century, the word was recorded in English through Latin and French, and later in the seventeenth century, it acquired its modern meaning [8]. Today, an idiom is generally understood as an expression whose figurative meaning differs from its literal meaning.

An idiom is a fixed expression composed of several words whose meaning cannot necessarily be understood when the words are translated individually [9]. What makes idioms interesting is that in earlier periods people used figurative language to express their feelings and describe situations. Hence, many idioms originated from everyday activities such as farming, trade, fishing, and hunting.

Challenges in translating idioms

Translating idioms presents several challenges.

- Context dependency.

  Different idioms have different contextual meanings depending on the situation. For example, the English idiom "throw in the towel" can be defined as abandoning a project, admitting defeat, or giving up in a competition. Therefore, it is necessary to understand the context and situation before deciding on the most accurate interpretation of the idiom.

- Non-literal meaning.

  Most idioms cannot be interpreted literally like ordinary vocabulary. They have meanings and interpretations that differ from a direct word-for-word translation. For example:

TABLE 2.1. EXAMPLES OF IDIOMS WITH THEIR LITERAL TRANSLATIONS AND ACTUAL MEANINGS.

Idiom	**Literal translation**	**Actual meaning**
Break the ice	Romper el hielo	Start a conversation
Spill the beans	Derramar los frijoles	Reveal a secret

- Difficulties for language learners.

  Language learners often fail to understand idioms and often mistake them for their literal meanings based on their native language. They may misunderstand the figurative meaning of idioms and, as a result, be unable to use them naturally.

- Cultural differences.

  This is the key aspect of the project, which compares different cultures to explain the same idiomatic meaning. Some English idioms have no direct equivalent in other languages, such as Spanish, even though they convey the same meaning. For example:

TABLE 2.2. EXAMPLE OF AN ENGLISH IDIOM AND ITS SPANISH EQUIVALENT.

Language	**Idiom**
English	It's raining cats and dogs
Spanish	Está lloviendo a cántaros

  Both expressions have the same meaning, which is heavy rain, but they use different imagery to convey it. In Spanish, it is not possible to say "Está lloviendo gatos y perros." instead of the appropriate Spanish idiom.

2.2. Neural Machine Translation

Neural Machine Translation (NMT) differs from traditional machine translation in its ability to jointly train a single neural network to optimize translation performance. This model follows an encoder-decoder architecture, where the encoder converts the source sentence into a fixed-length vector representation, and subsequently, the decoder generates the translation [10].

2.2.1. MarianMT Model

MarianMT model known as a machine translation model written in pure C++ with minimal dependencies, which allows for high translation speeds. A competent meta algorithm has been developed to train encoder-decoder models such as BART.

The Helsinki-NLP/opus-mt-en-es model was employed to produce natural Spanish translations of English idioms, allowing learners to compare the intended meaning of an idiom with its literal translation [11].

Furthermore, this model use encoder-decoder design, where the framework implements encoders and decoders as extensible classes. In this project, the encoder builds a context from English idiom as input, then passed to the decoder to produce natural

Spanish translation. This design also can be use to join different types of encoders and decoders, such as an RNN-based encoder with a Transformer decoder. As a result, this model able to maximize memory usage and achieve high translation speeds compared to traditional Python-based toolkits [12].

Figure 2.1 shows how to use Back Translation, a data augmentation technique but using French instead of Spanish. The original text stored in the JSON file was written in English and subsequently translated into Spanish using the MarianMT model.

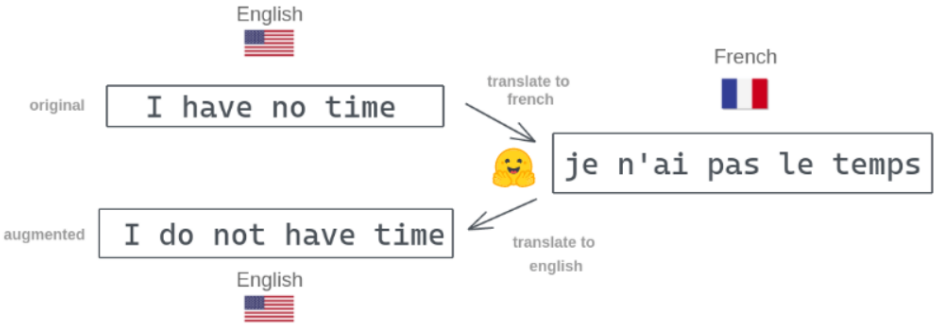


Fig. 2.1. Back Translation [13].

Data augmentation process works by installing Hugging Face transformers, then import MarianMT model and tokenizer.

Other Neural Machine Translation models that might offer good performance include the Google Translate API, and DeepL API. Google Translate API requires a paid subscription for usage exceeding 500,000 characters per month. It requires an internet connection and Google Cloud API credentials to make API calls. It is also a closed-source service managed through Google Cloud as a Software as a Service (SaaS) [14]. DeepL shares similar properties, as it requires a paid subscription, API access, and cloud infrastructure with ISO/IEC 27001 and System and Organization Controls 2 (SOC 2) enterprise security to operate [15]. In contrast, MarianMT is completely free, accessible without an API, and available as an open-source model [11]. Hence, MarianMT was chosen for this project because it is an open-source alternative that can be used without API credentials or subscription costs.

2.3. Large Language Models

Large Language Models (LLMs) are neural networks trained on large text corpora. They are most commonly built on the Transformer architecture, which is mainly used to predict and generate language [16]. The Transformer differs slightly from earlier recurrent architectures. It relies entirely on self-attention mechanisms, which allow it to model relationships between all tokens in parallel, whereas previous architectures processed them

sequentially. This design provided substantial improvements when scaling to larger models and datasets, leading to the development of the modern generation of LLMs.

Transformer-based models can be broadly categorized into two groups. Decoder-only, or causal, models are trained to predict the next token based on the preceding context, such as the GPT family of models [17], making them suitable for open-ended text generation. Meanwhile, encoder-decoder, or sequence-to-sequence, models are particularly suitable for tasks with a well-defined input-output relationship, such as instruction following or translation, because they process an entire input sequence before generating an output sequence, such as the T5 model [18]. This architecture was later extended by FLAN-T5 through instruction fine-tuning across diverse tasks, thereby improving its ability to follow natural-language prompts without additional task-specific training.

A wide range of natural language processing tasks make extensive use of transformer-based models. Different models may be better appropriate for particular applications based on their architectural features. For example, encoder-decoder models are commonly employed for translation, summarization, and instruction-following tasks, whereas decoder-only models are often preferred for open-ended text generation and conversational applications.

2.3.1. FLAN-T5-base

The Fine-tuned Language Net Text-to-Text Transfer Transformer (FLAN-T5) is structured as a text-to-text task that matches its encoder-decoder architecture and instruction-tuned framework.

The model was developed by Google and is available in several sizes with different numbers of parameters. It is pre-trained on the C4 (Colossal Clean Crawled Corpus), a large-scale dataset derived from web-crawled text [19]. The architecture of FLAN-T5 is based on the Transformer model, consisting of stacked transformer layers with self-attention and feed-forward sublayers, as shown in Figure 2.2 [20].

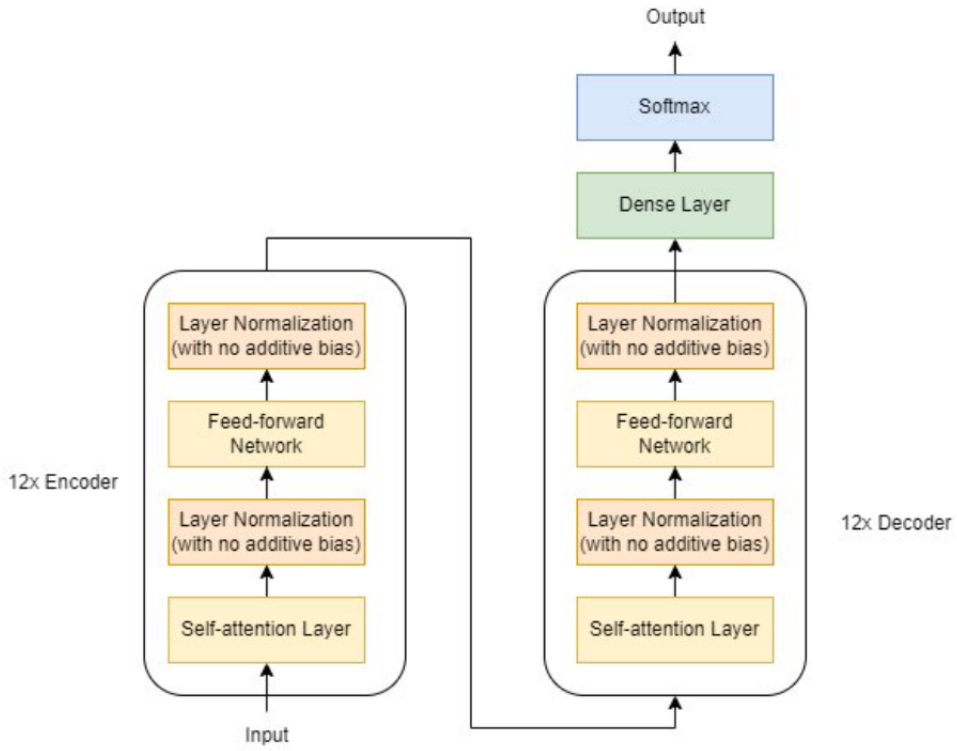


Fig. 2.2. FLAN-T5 architecture [20].

The diagram presents stacked transformer layers for both the encoder and decoder, where each consists of self-attention and feed-forward network (FFN) sublayers. However, each decoder layer also contains a cross-attention mechanism over the encoder output, positioned between self-attention and the FFN, while layer normalization is applied after each sublayer [21]. This structure allows the encoder to process all input tokens in parallel, since self-attention computes relationships between all token pairs simultaneously rather than sequentially. The decoder benefits from the same parallelism during training, but at inference it generates tokens autoregressively, one at a time, since each new token depends on those previously generated.

Since FLAN-T5 builds directly on the pretrained T5 model, it inherits two key architectural design choices: a relative position bias built into the attention mechanism, and a span-corruption objective used during pre-training. Rather than using fixed or absolute position embeddings, T5 adds a learned bias to attention scores based on the relative distance between tokens. This approach was adapted from Transformer-XL and was shown to better preserve sentence structure than absolute position encodings. During pre-training, T5 corrupts contiguous spans of text using mask tokens rather than masking individual words, and the model is trained to reconstruct the missing spans [22].

Building on this pretrained foundation, FLAN-T5 is further trained using instruction-prompting, allowing developers to perform specific tasks such as translation, classification, summarization [23], question answering [22], and other text-to-text tasks [19].

The model itself is available in five different sizes, as shown in Table 2.3. The total number of parameters increases with the size of the model, and each scale has a different T5 configuration [24].

TABLE 2.3. FLAN-T5 MODELS.

Model Variant	**Parameter Count**
FLAN-T5 Small	77M
FLAN-T5 Base	250M
FLAN-T5 Large	780M
FLAN-T5 XL	3B
FLAN-T5 XXL	11B

There are three other models that can be compared with FLAN-T5, namely DistilGPT-2, Mistral-7B, and GPT-4. DistilGPT-2 is an open-source model based on a pre-trained causal language model used for text generation and writing assistance. It can be run locally and consists of approximately 88.2 million parameters. However, after experimentation, its main drawback was the unreliability of the generated text, which often lacked consistency and quality [25].

Meanwhile, GPT-4 requires a paid API subscription for personal use while providing high-quality text generation capabilities. As a result, its integration may introduce additional operational costs and dependency on external services [26].

Mistral-7B, in contrast, is an open-source model consisting of 7 billion parameters. However, it is considerably larger and more computationally demanding, making it difficult to deploy under limited hardware resources [27].

Hence, FLAN-T5-base was selected for this undertaking since it strikes a good balance among the size of the model, its computational needs, and the quality of the generated text. Having around 250 million parameters, it is sufficiently light for use in environments with limited resources, while still delivering dependable abilities in following instructions and generating text.

2.3.2. Multilingual MiniLM

This model uses supervised or contrastive learning objectives to learn the alignment between meaning and vector space representations. For example, semantically similar sentences are brought closer together, whereas unrelated sentences are pushed further apart in the embedding space [28].

This sentence-transformer model maps sentences and paragraphs into a 384-dimensional dense vector space, which is useful for tasks such as semantic search and clustering [29]. In addition, the input text is limited to 256 words; otherwise, it is truncated. The resulting vector representation captures the semantic information contained in the text. The model

is initially trained through pre-training and later refined through fine-tuning using specific training data and hyperparameter settings. During fine-tuning, cosine similarities are computed between pairs of sentences within each batch [30].

Furthermore, the MiniLM architecture consists of 12 layers with 118 million parameters. It can capture semantic relationships by generating compact and meaningful embeddings through the application of a pooling mechanism to contextual representations of words. The model is also capable of generating sentence embeddings in multiple languages, which is particularly useful for understanding contexts in both English and Spanish [31].

There are several other models that provide similar functionality, such as paraphrase-xlm-r-multilingual-v1 and paraphrase-multilingual-mpnet-base-v2. A comparison between these models is summarized in Figure 2.3.

Modelo	Parámetros	Dimensiones de salida	Idiomas soportados	Rendimiento en STS-Benchmark
paraphrase-multilingual-MiniLM-L12-v2	118M	384	50	0.85
paraphrase-xlm-r-multilingual-v1	278M	768	50	0.87
paraphrase-multilingual-mpnet-base-v2	278M	768	50	0.88

Fig. 2.3. Comparison table of Multilingual Model [31].

In summary, the multilingual MiniLM-L12-v2 model provides competitive performance despite its relatively small size and number of parameters, making it an efficient choice for Natural Language Processing (NLP) tasks involving multilingual sentences.

2.4. Development Technologies

2.4.1. JSON

JSON: The Language of Data Exchange

JSON (JavaScript Object Notation) is a text-based, language-independent data format used in web APIs and machine learning pipelines. It has largely replaced XML as the primary medium for data exchange, due to its popularity and minimal syntax for representing various data structures [32]. Examples of such data structures include objects in JavaScript, maps in Java, and dictionaries in Python.

Based on technical structure and syntax, JSON is built on two universal structures:

- The Object: A collection of name or value pairs delimited in curly braces .
- The Array: An ordered list of values delimited in square brackets [] [33].

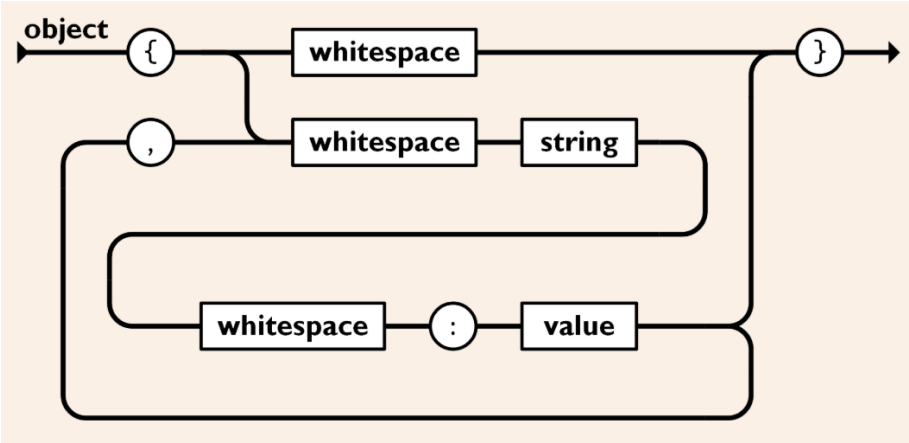


Fig. 2.4. JSON syntax diagram [33].

Figure 2.4 illustrates how the JSON object is contructed. The object pairs with curly braces { at both beginning and end of the structure. Within these braces, key-value pairs are defined, with a colon (:) and a matching value after each string key. Commas are used to separate multiple pairs of key-values, and an empty set of braces denotes an empty object.

JSON plays a crucial role in ensuring reproducibility in the machine learning pipeline. When a model is checkpointed, it is a directory which contains JSON files that define the logic of the weights instead of just a single file of weights.

There are two types of JSON file with different main functionalities:

- config.json: To define model architecture, such as dropout probabilities and number of attention heads.
- tokenizer.json: This file contains the mapping of sub-word tokens to numerical IDs, ensuring that during training and during inference the text is processed exactly the same way [34].

JSON primarily functions as the knowledge base and main source of information. It is known for its lightweight nature and human-readable format for data exchange [33]. Therefore, it is suitable for structuring and storing data efficiently. The format used is also very simple, as it is easy to process, read, and modify, and it can represent data using key-value pairs and hierarchical structures [34]. This format is commonly used with many programming languages due to its compatibility and flexibility [33], making it well suited for web applications and software development.

2.4.2. SQLite

SQLite is an in-process library that incorporates a zero-configuration, self-contained, transactional, and serverless SQL database engine. It is considered a very simple and easy-to-use database. Many projects, ranging from low-profile to high-profile applications, implement this database for a variety of purposes. Furthermore, SQLite is freely available in the public domain, allowing developers to easily access and use its source code [35].

However, SQLite is slightly different from traditional SQL database systems, as it is an embedded SQL database engine that does not require a separate server process. Instead, it has the ability to read and write directly to ordinary disk files. Hence, it has become a popular Application File Format, allowing developers to freely copy a database between big-endian and little-endian architectures, as well as between 32-bit and 64-bit systems [35].

The question that arises is: how does SQLite work?

- No server Architecture. SQLite is a compact and simple library that requires no additional installation.
- Data storage in a single file. Data is automatically stored locally in a single database file. Therefore, it can be easily moved and managed across different environments without complex procedures.
- Compatibility with SQL language. SQLite supports standard SQL syntax, making it easy to implement and use [36].

We can also highlight the main characteristics of SQLite:

- Lightweight engine with low resource consumption.
- Compatibility with standard SQL.
- High portability between operating systems.
- Minimize configuration and easy implementation.
- Optimized performance for local applications [36].

Therefore, SQLite is mainly used in web development, mobile applications, and desktop applications.

2.4.3. Streamlit

Streamlit is an open-source framework specially designed for web development. This Python-based library provides an interactive web-based learning platform without requiring traditional front-end technologies such as HTML, JavaScript, or CSS. With only a few

lines of code, developers can transform project code into a visually appealing application [37].

Furthermore, Streamlit provides an intuitive and effective user interface with simpler development steps, enabling developers to build interactive user interfaces in a short amount of time.

There are several characteristics of Streamlit that can be taken into account:

- Interactive web applications can be created with less code and development time.
- Compatible with most Python libraries such as Pandas, Seaborn, PyTorch, Matplotlib, and Keras.
- No front-end (HTML, JavaScript, or CSS) knowledge or experience is required.
- Data caching simplifies and speeds up computation pipelines [37].

Streamlit is built based on a client–server structure where users access the application through a Python backend that serves as the server, while the client is the frontend view displayed in a browser. Both the server and client run locally once the application is developed on a computer. To start a Streamlit server on a computer with Python installed, the command `streamlit run your_app.py` must be executed. Once the server is started, it handles all calculations for users who visit the application. The machine hosting the application is called the host. Besides, the Streamlit client is the device through which the user accesses the application, also known as the browser frontend [38].

Additionally, Streamlit has an integrated caching system that only runs a function once at first. Instead of recalculating the function, the cached result is returned during consecutive runs. Every time the function code is changed, the cache is automatically updated.

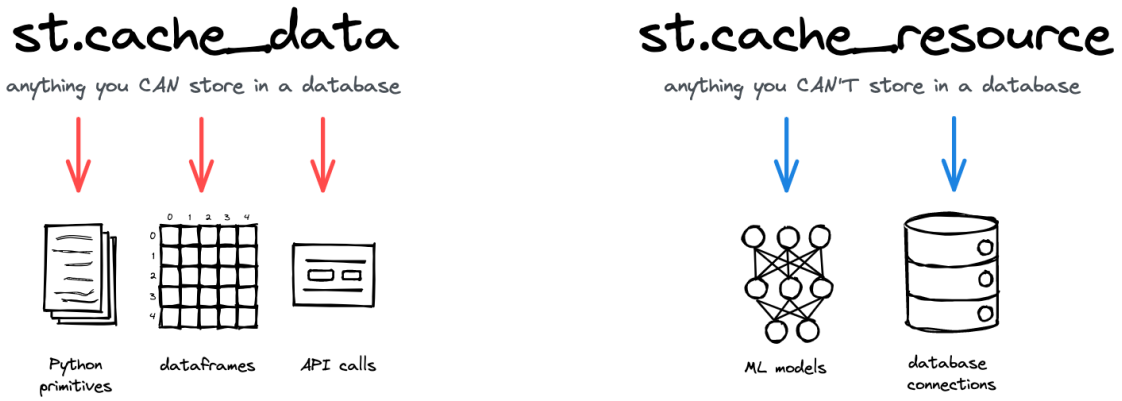


Fig. 2.5. Caching diagram [39].

Figure 2.5 illustrates the two caching decorators, `st.cache_data` and `st.cache_resource`. Whenever a task involving data retrieval is required, `st.cache_data` is used to cache

computations such as loading a DataFrame from a CSV file, querying an API, transforming a NumPy array, or functions that return serializable data objects such as arrays, lists, integers, floats, and strings. In contrast, `st.cache_resource` is mainly used to cache global resources such as database connections or machine learning models. These are objects that are not serializable and should not be loaded repeatedly. It allows these resources to be shared without replicating or duplicating them across multiple reruns and user sessions of an application. It should be noted that modifying the cached return value also changes the cached object itself [39].

2.4.4. Hugging Face

The Hugging Face: The "Model Hub" Revolution.

Hugging Face was founded as a chatbot company, emerged as the definitive platform for Open Source AI. This kind of infrastructure allows researchers and developers to move from specific concepts to inference in minutes instead of usual case which is in weeks.

The hugging face hub consists of a centralized repository system powered by Git-LFS (Large File Storage). It stores three primary types of assets:

- Models: Pre-trained weights for NLP, audio, Computer Vision and Multimodal tasks.
- Datasets: Massive collections of raw data formatted for immediate use such as text, images, or audio.
- Spaces: Interactive web applications that allow users to demo models directly in the web browsers. Web applications offered by Hugging Face such as static HTML, Streamlit, and Gradio.

The transformers library provides a high-level API to download and train cutting-edge models. It is designed with a low barrier to entry while still offering a high ceiling for researchers which manages the two-step process of AI interaction:

- Tokenization: Using Tokenizer to convert raw text into numbers.
- Forward Pass: In order to get prediction, passing those numbers through the neural network.

In addition to models, Hugging Face offers specialized tools:

- tokenizers: Normally written in rust for extreme speed, it handles the conversion of strings to integers.

- datasets: Data prevents clogging the memory by load a 100GB dataset on a laptop with only 8GB of RAM as the data is read directly from the disk as needed.This allows solving of “Big Data” problem by using Apache Arrow memory mapping [40].

2.4.5. Google Text-to-Speech (gTTS)

gTTS is a simple and effective API that is commonly used in educational demonstrations, language-learning systems, and rapid prototyping environments.

The audio is generated on Google’s servers. Therefore, an internet connection is required to ensure that this functionality operates correctly. Initially, a request is sent to the backend service associated with the Google Translate "Listen" feature. Subsequently, the generated MP3 file is downloaded and delivered to the application. As a result, the generated speech can be played within software applications or exported as audio files for later use [41].

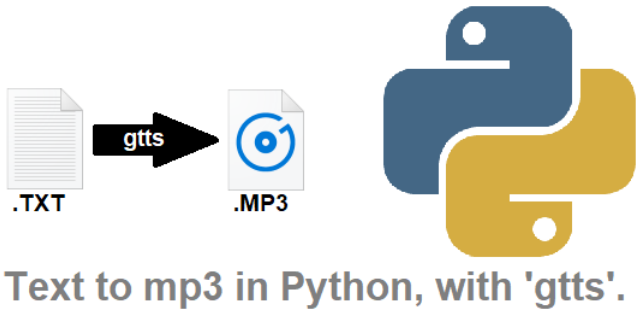


Fig. 2.6. gTTS architecture. Source: Adapted from [42].

gTTS integrates well with Python workflows and provides functionalities such as MP3 generation, file saving, and output streaming through standard output (stdout), making it suitable for scripting and automation purposes. Furthermore, it supports approximately 60 languages and numerous regional dialects through language and top-level domain (TLD) parameters [43].

This technology can support vocabulary acquisition and retention through a dual-modality learning approach. Dual Coding Theory, which has been shown to enhance vocabulary recall among second-language learners, states that information is processed more effectively when presented through both verbal and non-verbal channels simultaneously. Specifically, combining written and auditory information can enhance learning [44]. Samburskiy (2020) reported a positive effect of a dual-coding approach on idiom interpretation among ESL/EFL learners [45]. This effect has been particularly investigated in the context of idiomatic language. More generally, text-to-speech technology has been found to be a useful tool for improving pronunciation awareness, listening practice, and

vocabulary development by reinforcing the connection between written and spoken word forms [46]. Therefore, learners' memory can be strengthened by pairing written words or expressions with their corresponding audio pronunciations, helping learners acquire correct pronunciation while supporting memorization.

Forvo is another audio feature that performs very well but has several limitations compared to gTTS. For example, it does not offer an audio caching mechanism, has an expiration period of only about two hours, requires a paid subscription for high usage, limits requests and available words, and has API restrictions [47]. Therefore, gTTS was chosen for this project due to its properties, including zero cost, high scalability, free caching for storage, and no API requirement.

2.5. Language Learning Tools

2.5.1. Duolingo

Duolingo is the world's most popular educational language-learning platform, founded by Luis von Ahn (a Carnegie Mellon University professor) and his postgraduate student, Severin Hacker, in 2011 [48]. It operates on a freemium business model [49] and offers courses in 42 languages, along with additional subjects such as music, mathematics, and chess. In 2020, Duolingo also released Duolingo ABC, designed by education experts to help children learn to read.

This application adopts a gamification-based approach where users can track their progress, maintain a streak, and compete with other players globally through various medals and achievements [50]. In addition, Duolingo uses an evaluation system, which encourages users to continue learning until they achieve specific objectives, such as earning a certain amount of Experience Points (XP). XP is a measurement unit used in role-playing video games and tabletop role-playing games (RPGs) to measure the experience and progression of a player's character throughout the game, including after completing objectives, participating in role-playing activities, and overcoming obstacles and rivals [51].

Users around the world can compete against each other in different leagues and have the opportunity to advance to higher leagues each week by gaining XP. The leagues in Duolingo, from lowest to highest, are Bronze, Silver, Gold, Sapphire, Ruby, Emerald, Amethyst, Pearl, Obsidian, and Diamond [52]. The system of this application is also based on spaced repetition to promote memorization in language learning [53].

As a result, users can more easily remember vocabulary due to their engagement with the course material. The repetitive lessons provided by Duolingo require consistent daily practice to progress and achieve the desired level. It also offers a learning mechanism with personalized recommendations and feedback.

Furthermore, some of its courses are aligned with the CEFR, including the most-

learned languages English, French, and Spanish. The Duolingo score ranges from 0 to 130 and is aligned with CEFR levels from beginner level A1 to B1 [50]. Additionally, users can add their scores to LinkedIn for career development [54].

Starting in 2024, Duolingo also added AI features to its platform to explain users' mistakes [55].



Fig. 2.7. Duolingo app. Source: Adapted from [56].

2.5.2. Memrise

It was launched in 2010 and founded by Ben Whately, Greg Detre (a Princeton neuroscientist), and Ed Cooke (a Grand Master of Memory) as a British language-learning app based in London, United Kingdom [57]. It was once announced as the "Best App" at the second edition of the Google Play Awards and is available for users to download on both Google Play and the App Store [58].

The concept of this platform is based on spaced repetition, which is known as one of the key techniques for memorization in language learning [59]. Apart from that, it also offers learning in a gamified way, similar to Duolingo, where users can track their progress through score charts.

The platform offers more than 140 languages and had over 87 million registered users as of 2026 [57]. Apart from language learning, Memrise also offered other subjects such as history, mathematics, geography, and natural sciences for users to practice their interests [60]. However, these subjects were moved to separate sites in 2024 [61].

What makes this app different from other language-learning apps is its community course feature, which offers community-created courses developed by users [62]. Therefore, users can focus on specific themes such as slang, medical terms, travel essentials, and mild swearing [63]. This feature is very helpful for those who want to improve their skills in these specific areas.



Fig. 2.8. Memrise app. Source: Adapted from [64].

2.5.3. Busuu

Busuu was founded by Bernhard Niesner and Adrian Hilti in May 2008. It initially only offered a free version on its website and later developed a premium membership where premium members had access to all features on the platform. Then, it launched its first mobile application in 2010 [65].

The main distinguishing feature of this application is that it shows users visually how to pronounce words and sentences through short videos of native speakers in real locations. Other native speakers can also correct and help users virtually by writing messages in forums and sending audio recordings [66].

This application consists of 14 language courses, taught in 15 interface languages, and provides learning materials based on international CEFR levels A1, A2, B1, B2, and C1 [67]. The main content of this app focuses on pronunciation exercises and conversational practice with native speakers who are also members of the Busuu community [66]. However, it also offers interactive quizzes and helps learners with vocabulary and grammar practice.

Furthermore, although users can learn freely anytime without restrictions, they also have the opportunity to obtain a level completion certificate in certain languages such as Spanish, English, Italian, Portuguese, French, and German [68].



Fig. 2.9. Busuu app. Source: Adapted from [69].

2.6. Contribution of This Work

Unlike general-purpose computer-assisted language learning (CALL) platforms, this project specifically targets bilingual idiomatic competence. It is an idiom-learning application rather than a typical language-learning application. It allows users to focus on differentiating idioms used in two different languages, such as the English–Spanish idioms implemented in this project. A comparison can be made between this project and other language-learning platforms that provide vocabulary lessons, grammar exercises, pronunciation practice, and reading-based activities, such as Duolingo, Busuu, and Memrise. For example, Memrise has already introduced an idiom section in one of its word lists to reinforce learning; however, it does not focus on differentiating idioms between two languages. Instead, it mainly presents common idioms used by native speakers through its premium version. One feature that clearly differentiates this project is that users can directly understand and use the equivalent term, slang, or idiom in the target language. This is particularly helpful for English speakers learning Spanish, allowing them to use the language more comfortably in everyday situations. Spanish speakers can also learn the usage of English idioms more effectively.

The application provides idiomatic counterparts whenever possible, in contrast to traditional translation tools that frequently provide literal translations that might not maintain the original figurative meaning. This enables learners to understand both the meaning of a phrase and how native speakers naturally express the same concept in a different language. Therefore, they can speak the target language in a more genuine and culturally appropriate manner.

A feature that distinguishes this application from previous tools is the way AI integration techniques are provided throughout the learning process. Duolingo Max, for example, uses OpenAI's GPT-4 as an LLM provider to include AI-based conversations through video calls [70], grammar explanations, and adaptive difficulty through the BirdBrain machine-learning engine [71]. Memrise also uses a similar LLM provider to provide MemBot as an interactive conversational AI with users, along with contextual feedback and spaced-repetition adaptivity [72]. Busuu, on the other hand, incorporates fine-tuned LLMs to provide conversational AI, post-check feedback, and mistake-repair sessions based on user error logs [73]. The new approach in this project uses a tailored pipeline for cross-linguistic idiom mapping and semantic retrieval, whereas many commercial solutions use LLMs for general grammar and broad conversation.

MarianMT is used as a translation tool to generate literal Spanish translations of the idioms, helping users compare literal and figurative meanings. Next, FLAN-T5 is implemented in the application to generate contextualized quiz questions dynamically, providing a variety of practice opportunities for learners rather than relying only on preset tasks. Furthermore, Multilingual MiniLM allows the recognition of idiomatic expressions even when they are embedded within longer sentences by enabling semantic idiom detection from free-text user input. When compared to conventional flashcard-based solutions, this

offers a more interactive learning experience.

Additionally, the application employs audio pronunciation through gTTS as a secondary form of learning alongside written forms. Moreover, SQLite is incorporated to enable users to track their performance and identify areas that require more practice through learning analytics and personalized progress tracking.

Mastering idiomatic expressions is often considered one of the final stages of language proficiency. While learners may acquire grammar and vocabulary relatively quickly, understanding idioms requires not only linguistic competence but also cultural awareness and contextual knowledge. Therefore, even advanced language users may struggle when faced with unfamiliar local idioms in daily interactions. Thus, this project aims to address a learning area that is frequently overlooked by current language-learning platforms by integrating bilingual idiom mapping, artificial intelligence, pronunciation support, and adaptive practice activities within a single platform.

3. METHODOLOGY

3.1. Dataset Collection

In this project, datasets were obtained from different sources. The idiom dataset was taken from `idiom2.json` (see Appendix A for the complete dataset), which was later integrated into the Hugging Face Space. Meanwhile, two example sentence datasets containing idioms were obtained from Hugging Face as explained later in Section 3.1.2 and were available for direct integration into the application.

In the application, four distinct components of the Hugging Face ecosystem have been utilized:

- **Hugging Face Hub:** `load_ translation_model()` function pulls the `Helsinki -NLP/opus-mt-en-es`[1] model.
- **Hugging Face Datasets:** Two datasets, `fdelucaf/IdioTS`[2] and `UCSC-Admire /idiom-SFT-dataset`[3] were used to extract examples of the idioms using function `build_examples_map()`. This allows the application to automatically demonstrate real-world usage examples.
- **Transformers Pipeline**:The Transformers library is used to load and execute pre-trained models through the `pipeline()` interface. In this project, the `google/flan -t5-base`[4] model is used for AI-powered quiz generation, enabling the creation of dynamically contextualized sentence-completion questions rather than relying solely on predefined examples.
- **Sentence Transformers:** Semantic idiom detection is done using the `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`[5] model. The program can recognize semantically related idiomatic expressions even when users do not enter the specific idiom since it transforms user input and idiom descriptions into vector embeddings.

[1] https://huggingface.co/Helsinki-NLP/opus-mt-en-es
[2] https://huggingface.co/datasets/fdelucaf/IdioTS
[3] https://huggingface.co/datasets/UCSC-Admire/idiom-SFT-dataset-561-2024-12-06_00-40-30
[4] https://huggingface.co/google/flan-t5-base
[5] https://huggingface.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2

3.1.1. Idiom Dataset

`idiom2.json`

The idiom dataset was created manually in JSON format to store the idioms used by the application. The dataset serves as the main knowledge base of the project and supports its purpose as an application to learn English-Spanish idioms. Provides the initial content of the prototype, which means that users can learn only the idioms included in this dataset in this early version of the application.

The file used consists of 119 different english idioms with their equivalent idioms in spanish. It acts as the main source of the application with the `load_idioms()` function, parses this file into a Python dictionary, allowing O(1) lookup time when a user selects a specific phrase. Also used within Streamlit's session state in order to track user XP and quiz progress, enabling the application to stay reactive and maintains state across interactions.

Table 3.1 below shows examples of the content of the dataset.

TABLE 3.1. SAMPLE ENTRIES FROM THE IDIOM DATASET (`IDIOM2.JSON`), SHOWING IDIOM, SPANISH MEANING, AND TOPIC CATEGORY.

English Idiom	**Spanish Equivalent**	**English Meaning**	**Topic**
Break the ice	Hacer que la gente se sienta cómoda	To do or say something to relieve tension or awkwardness, especially when meeting someone new	Communication
Step up to the plate	Dar un paso al frente / Asumir la responsabilidad	To take responsibility or action when needed.	Responsibility
Off the beaten track	Fuera de lo común / En un lugar remoto	In a remote or unusual place, away from popular routes	Travel

The idiom dataset contains 119 idioms spanning 32 thematic categories (e.g. Communication, Money, Health, Relationships), each entry consisting of an idiom, its natural Spanish meaning, English meaning, and an assigned topic used for filtering in the application. A representative sample is shown in Table 3.1 and Appendix A for the full idiom dataset.

Topic categorization was added to the dataset to allow users to browse idioms by specific topics based on their learning interests. These thematic categories are Animals, Awareness, Challenges, Commitment, Communication, Creativity Daily Life, Decision

Making, Dreams, Efficiency, Emotions, Health, Hobbies, Humor, Judgment, Leadership, Life & Death, Luck, Memory, Money, Motivation, Optimism, Patience, Personality, Problems, Relationships, Responsibility, Risk, Secrets, Society, Success & Failure, Time, Travel, Work.

3.1.2. Example Sentences Dataset

The IdioTS dataset provides bilingual examples of idiomatic expressions and is available on Hugging Face. Hence, this dataset was incorporated into the application to provide English and Spanish sentences containing idioms in context. These examples were integrated to illustrate authentic usage and are displayed in most of the application modes.

The UCSC-Admire dataset contains literal and idiomatic example sentences with selected idioms, and is also associated with images related to the examples. This dataset was used to provide additional examples of idioms that were not present in the IdioTS dataset. Therefore, users can enhance their learning experience through a wider variety of idiom examples in different contexts. As discussed earlier, real examples are one of the most effective methods for understanding and remembering the meaning and usage of idioms.

3.2. Data Processing

3.2.1. Data cleaning

Several preprocessing steps were performed before integrating the datasets into the application to ensure usability and consistency of the data. The multiple datasets retrieved from Hugging Face required normalization before use, as they contained idiomatic expressions and example sentences in different formats.

Primarily, all idiom entries were normalized by converting them to lowercase to ensure consistency during search operations and idiom detection. Leading and trailing whitespace characters were removed to maintain consistent formatting and prevent duplicate idiom entries.

In addition, null entries were handled by filtering out missing or empty values whenever possible. This cleaning process was very important to ensure that only valid English and Spanish example texts were retained, while at the same time reducing potential errors during retrieval and ensuring that only relevant examples were presented to users.

As a result, the cleaned data, with standardized formatting of idioms and example sentences, facilitates efficient lookup and integration within the application.

3.2.2. Building the Examples Map

The next step after data cleaning was organizing the example sentences into an examples map, a dictionary structure. In this structure, each idiom is associated with a collection of corresponding English and Spanish example sentences constructed from two different datasets: IdioTS and UCSC-Admire. For each record, the idiom serves as the dictionary key, while the associated example sentences are stored as the corresponding values. This allows a single idiom to be linked to multiple examples.

An example representation is shown below:

```
hold your horses
  English: Hold your horses! There's no need to rush into it right now,
           just take a day or two to think it through.
  Spanish: Para el carro! No hace falta precipitarse ahora,
           tómate uno o dos días para pensártelo bien.

break a leg
  English: You're going to do great at that speaking event,
           so don't worry and break a leg!
  Spanish: Lo vas a hacer muy bien en la conferencia,
           así que no te preocupes y mucha suerte!
```

This mapping structure enables efficient retrieval of example sentences during application runtime. When a user selects an idiom, the corresponding examples can be accessed directly through the dictionary key without repeatedly searching the original datasets. Consequently, the examples map enhances both the performance and maintainability of the system.

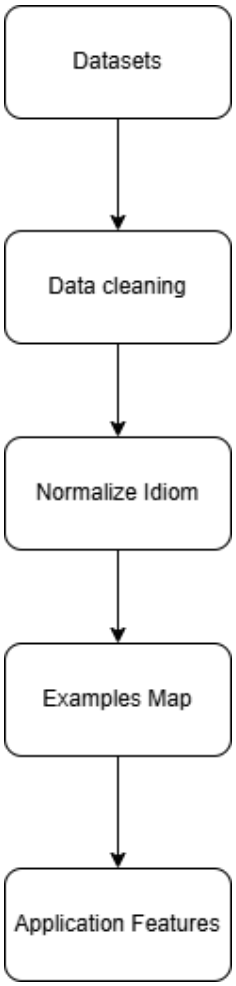


Fig. 3.1. Workflow for constructing the examples map from the collected idiom datasets.

Figure 3.1 illustrates the process used to construct the examples map. After the datasets were collected, data cleaning and normalization procedures were carried out before structuring the examples into a dictionary that associates each idiom with its corresponding English and Spanish example sentences.

3.3. System Architecture

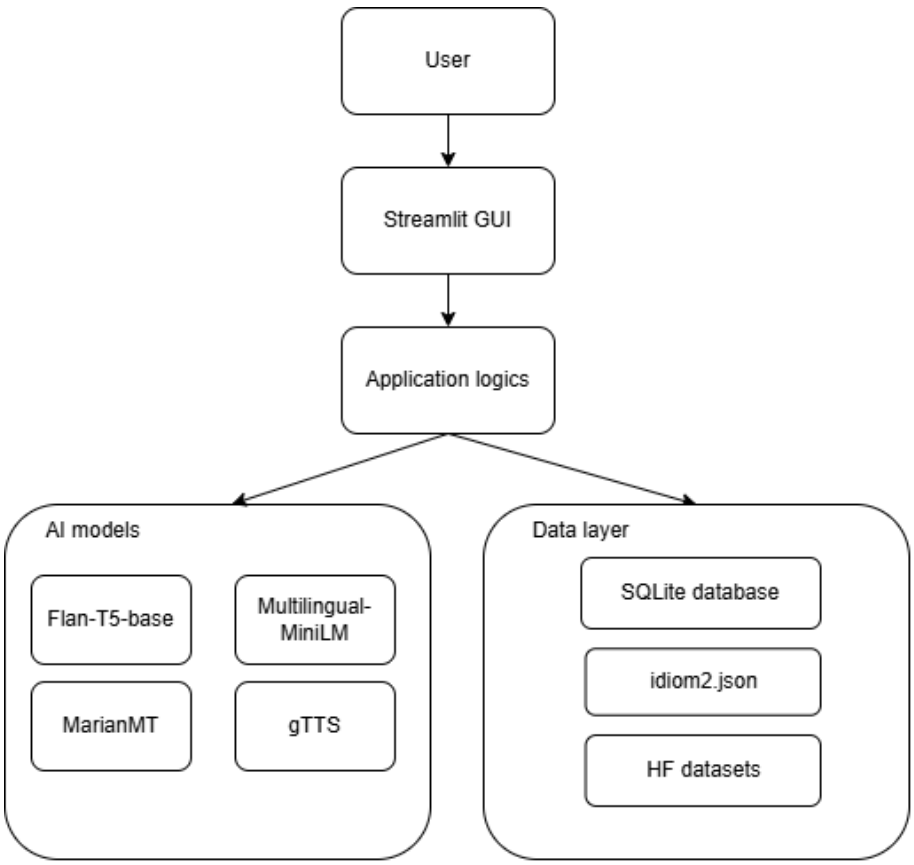


Fig. 3.2. Overall architecture diagram.

Figure 3.2 illustrates the overall architecture of the proposed application. To begin with, users have access to the application through a web browser via a Streamlit-based graphical interface. The Streamlit User Interface (UI), defined in `app.py`, allows users to freely navigate through multiple modes, such as Explore, Quiz, and Analytics. The application logic `utils.py` is where the models and data are initially orchestrated. Subsequently, the application logic relies on two main components: the AI models and the data layer. The AI models used in this application are FLAN-T5-base, which generates quiz sentences; Multilingual-MiniLM, which performs idiom detection; MarianMT, which provides literal English–Spanish translations; and gTTS, which enables audio pronunciation. Meanwhile, the data layer consists of an SQLite database to store favorites and the learning analytics table, `idiom2.json`, which provides the base set of idioms, and the Hugging Face datasets, which provide additional example sentences for certain idioms. The entire source code of the application, including `app.py`, `utils.py`, and `idiom2.json`, is available in the project's GitHub repository[6].

[6] https://github.com/kawaa099-dot/english-spanish-idioms-app

3.4. Back-End Components

This section explains the back-end components integrated into the final implementation of the application. These technologies manage the application logic, data storage, and Artificial Intelligence (AI)-driven functionalities.

3.4.1. Streamlit

The application is built using Streamlit as its primary framework, which manages the graphical user interface, front-end and back-end integration, and user interactions. Users are able to interact with different learning modes through Streamlit components such as buttons, select boxes, text boxes, progress bars, and charts [37]. Additionally, Streamlit Session State is utilized to preserve quiz progress, XP scores, and adaptive learning data throughout a user session.

3.4.2. SQLite

SQLite is used as the main storage system to locally store user-related information. It contains two primary tables that are discussed later in Section 4.1.4: the favorite table and the analytics table. The favorite table stores idioms saved by users, while the analytics table records the number of attempts and correct responses for each user. This information enables the application to generate learning metrics and supports the personalized learning process by identifying weak idioms that require additional practice.

3.4.3. gTTS

The audio generation process can be summarized in three main steps:

1. The selected idiom is converted into a filename and associated with an MP3 audio file.
2. The local cache is checked to determine whether an audio file already exists for the selected idiom.
3. If no audio file is found, gTTS generates the speech output and stores it locally for future use.

This caching mechanism prevents repeated audio generation for the same idiom requests, reducing processing time and improving overall application performance. Consequently, users experience faster audio playback since previously generated audio files can be reused immediately.

Despite its advantages, gTTS has several limitations, including the lack of highly natural intonation and limited voice customization options. However, these limitations do not significantly affect the objectives of this application, as audio pronunciation serves only as a supplementary feature rather than the primary focus of the system [41].

3.5. AI Integration

This section discusses the integration of selected pre-trained language models into the application. The application performs certain NLP tasks, such as literal translation, quiz creation, and idiom detection, using publicly available models from Hugging Face. Each model was chosen on the basis of how well it fit the corresponding functionality.

Several pre-trained models obtained from Hugging Face provide AI-powered functionalities within the application. The MarianMT model was integrated to generate literal English-to-Spanish translations of idioms, while FLAN-T5-base was used to generate AI-powered quiz questions for the Quiz mode. Last but not least, the sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2 model was employed for semantic idiom detection by measuring the cosine similarity between sentence embeddings.

3.5.1. MarianMT for Translation

MarianMT was integrated to provide literal English-Spanish translations of idiomatic expressions in the mode Explore Idioms of the application. When a user chooses an idiom, the application sends the text to the MarianMT model, which literally translates it into Spanish. The definition of the idiom, sample sentences, and audio pronunciation are then shown alongside the translated output.



Fig. 3.3. Flowchart MarianMT diagram.

3.5.2. FLAN-T5-base for Quiz Generation

In this project, FLAN-T5-base was selected because it better fits the available computational resources (CPU and RAM) provided by Hugging Face Spaces. FLAN-T5-base consists of approximately 250 million parameters and uses 12 transformer layers in both the encoder and decoder.

FLAN-T5-base is employed to create structured outputs that follow predefined criteria rather than generating new quiz sentences entirely from scratch, which is not always reliable. The model obtains genuine idiomatic example sentences sourced from the combined datasets and rewrites them while maintaining the original figurative meaning and preserving the intended idiom. This method combines the reliability of well-structured educational materials with the flexibility of AI-generated content, resulting in a greater diversity of quiz questions.

Figure 3.4 illustrates the workflow of AI-generated quiz questions as implemented in the application. Initially, a random idiom is selected from the idiom repository, and the associated example sentences are retrieved. The sentence is then passed to the FLAN-T5-base model as input.

The model is instructed to paraphrase the sentence while maintaining its original meaning and preserving the idiomatic expression. Several validation steps are subsequently performed to ensure the quality of the generated output. These include checking whether the idiom is preserved, ensuring that the generated sentence is different from the original sentence, avoiding repetitive outputs, verifying that the sentence does not exceed the maximum length limit of eight words as defined in this project, and preventing prompt leakage. After that, the idiom is replaced with a blank space.

Next, three distractors consisting of different idioms are generated to satisfy the multiple-choice question format before the complete quiz question is displayed in the Quiz Time mode.

Otherwise, if the generated output is found to be invalid, the application retries the AI generation process, and the user is required to select the "New Question" button again. Based on the number of attempts, which is configured as a maximum of five in this application, the system continues generating new outputs while attempts remain available. However, if the maximum number of attempts is exceeded, the application falls back to the example sentence dataset.

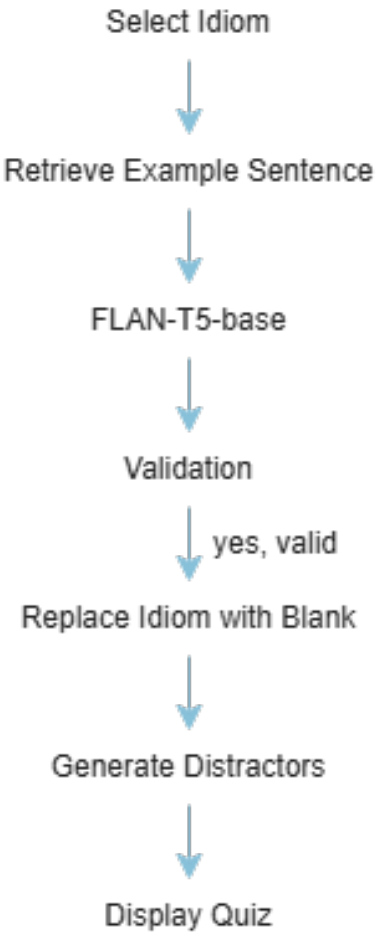


Fig. 3.4. Flowchart FLAN-T5-base diagram.

There are three other models that can be compared with FLAN-T5, namely DistilGPT-2, Mistral-7B, and GPT-4. DistilGPT-2 is an open-source model based on a pre-trained causal language model used for text generation and writing assistance. It can be run locally and consists of approximately 88.2 million parameters. However, after experimentation, its main drawback was the unreliability of the generated text, which often lacked consistency and quality [25].

Meanwhile, GPT-4 requires a paid API subscription for personal use while providing high-quality text generation capabilities. As a result, its integration may introduce additional operational costs and dependency on external services [26].

Mistral-7B, in contrast, is an open-source model consisting of 7 billion parameters. However, it is considerably larger and more computationally demanding, making it difficult to deploy under limited hardware resources [27].

Hence, FLAN-T5-base was selected for this undertaking since it strikes a good balance among the size of the model, its computational needs, and the quality of the generated text. Having around 250 million parameters, it is sufficiently light for use in environments with limited resources, while still delivering dependable abilities in following

instructions and generating text.

3.5.3. Multilingual MiniLM for Semantic Idiom Detection

In the Idiom in Sentences mode, the application looks for idiomatic expressions in user-provided text. While more details are retrieved and shown, such as meaning, literal translation, pronunciation, and sample sentences, the detected idioms are highlighted within the sentence. Therefore, users can learn idioms in a more proactive way.

This application integrates the `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` model to support idiom detection from free-text sentences entered by users in the Idioms in Sentences mode.

Initially, the `all-MiniLM-L12-v2` model was considered, but it has limited robustness for languages other than English. Therefore, the multilingual embedding version was used instead to support multiple languages. As a result, users can provide either English or Spanish sentences as input text.

Figure 3.5 below illustrates the workflow of Multilingual MiniLM. When a user enters a free-text sentence, the text is first converted into a sentence embedding, which is a numerical representation generated by the Sentence-Transformers model. Next, cosine similarity is used to measure the semantic closeness between the embedding of the user input and the embeddings of idiom examples stored in the dataset. The idiom associated with the most similar example sentence is then selected as the detected idiom. Finally, the application retrieves and displays the detected idiom together with its meaning, literal translation, pronunciation audio, and example sentences.

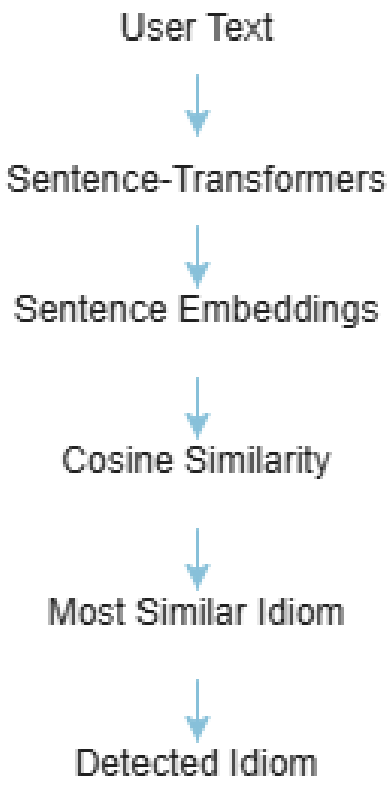


Fig. 3.5. Flowchart Multilingual MiniLM diagram.

4. SYSTEM IMPLEMENTATION

4.1. Application Architecture

The application architecture was previously presented in Figure 3.2 in Section 3.3. Users interact with the application through the Streamlit-based interface implemented in `app.py`, which communicates with backend functions defined in `utils.py`. This section describes the implementation and integration of these components within the application.

4.1.1. `app.py`

The Python file `app.py`[7] plays a main role in the application, where the user interface is implemented using Streamlit. Here, users are able to navigate between learning modes and interact with the backend functions developed in another Python file, `utils.py`.

The navigation system to switch between the learning modes of the application is shown in Listing 4.1.

CÓDIGO 4.1. Navigation menu implementation using Streamlit.

```
mode = st.sidebar.selectbox(
    "Choose a mode",
    ["Explore Idioms", "Idioms in sentences",
     "Quiz time!", "Favorites",
     "Learning Analytics"]
)
```

4.1.2. `utils.py`

The `utils.py`, located in the same repository, backend module consists of the core application logic, such as dataset processing, translation, audio generation, quiz generation, idiom detection, and database management.

CÓDIGO 4.2. Function used to build the examples map.

```
def add_example(key, en, es):
    key = key.lower().strip()

    if key not in examples_map:
        examples_map[key] = []

    if en or es:
```

[7] https://github.com/kawaa099-dot/english-spanish-idioms-app

```
            examples_map[key].append({
                "en": en or "",
                "es": es or ""
            })
```

Listing 4.2 illustrates the examples map construction used to provide available examples for idioms.

As the example datasets from Hugging Face were integrated into the application, they first needed to be organized into a dictionary structure to improve retrieval efficiency. The `add_example()` function was implemented to associate each idiom, or key, with its corresponding English and Spanish example sentences. In the second line of the function, the idiom is normalized by converting it to lowercase and removing unnecessary whitespace to avoid duplicate entries and inconsistent storage.

If an idiom does not exist in the dictionary, a new entry is created, where the English and Spanish example sentences are stored as values associated with that idiom. Consequently, multiple example sentence pairs are stored as a list of values, while each idiom functions as a dictionary key. This structure enhances the maintainability of the dataset and allows quick retrieval of examples during application execution.

4.1.3. `idiom2.json`

The idiom repository includes pairs of idiomatic expressions and their definitions, and it is stored in JSON format.

CÓDIGO 4.3. Example of idiom data stored in JSON format.

```
{
  "spill the beans": "reveal a secret",
  "break the ice": "initiate conversation"
}
```

Listing 4.3 shows an example of how idioms and their meanings are stored in JSON format. Each idiom is represented as a key, and the associated value stores the meaning of the idiom.

Listing 4.4 further shows the modification made to include the topic for browsing idioms and the English meaning for better comprehension.

CÓDIGO 4.4. New example of idiom data stored in JSON format.

```
{
  "break the ice": {
    "meaning": "hacer que la gente se siente cómoda",
    "topic": "Communication",
```

```
    "english_meaning": "To do or say something to relieve
       tension or awkwardness, especially when meeting someone
        new."
  },
  "hit the sack": {
    "meaning": "irse a la cama",
    "topic": "Daily Life",
    "english_meaning": "To go to bed or go to sleep."
  }
}
```

4.1.4. SQLite Database

The SQLite database was chosen for this project due to its lightweight architecture, simple Python integration, and suitability for web-based and desktop applications that do not require a dedicated database server. SQLite is used in the proposed application to store user-related data, enabling the maintenance of learning progress and preferences between sessions.

In this project, it is mainly used in Modes 4 and 5, where SQLite is integrated with Python applications. The database stores persistent user information and learning progress without requiring an external database server.

- Mode 4: Favorites.

CÓDIGO 4.5. Creating the favorite table in SQLite.

```
CREATE TABLE IF NOT EXISTS favorite (
    idiom TEXT PRIMARY KEY,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

The Listing 4.5 shown demonstrates how the table is created, together with its associated functions: `add_favorite()`, `get_favorite()`, and `remove_favorite()`.

The main objective of these functions is to store idioms marked by users for later review.

TABLE 4.1. STRUCTURE OF THE FAVORITE TABLE.

Field	**Type**
idiom	TEXT
timestamp	DATETIME

The favorite table allows users to select idioms and save them as favorites for future

reference. Therefore, it enables users to create a personalized collection of idioms that they find difficult for further learning or that simply catch their attention.

As the primary key, the idiom field prevents the same idiom from being added more than once. The timestamp field records the date and time when the idiom was added to the favorites list.

Users can later access the Favorites mode, where all the contents of the table are retrieved, including the saved idioms and their meanings.

- Mode 5: Learning analytics.

CÓDIGO 4.6. Creating the analytics table in SQLite.

```
CREATE TABLE IF NOT EXISTS analytics (
    idiom TEXT PRIMARY KEY,
    attempts INTEGER DEFAULT 0,
    correct INTEGER DEFAULT 0
);
```

The Listing 4.6 shown demonstrates how the table is created, together with its associated functions: `update_analytics()`, `get_learning_stats()`, and `get_weak_idioms()`.

By combining Mode 3 (Quiz) and Mode 5 (Learning Analytics), users are able to view the number of attempts, number of correct answers, accuracy for each idiom, weak idiom detection, and adaptive quiz recommendations.

TABLE 4.2. STRUCTURE OF THE ANALYTICS TABLE.

Field	**Type**
idiom	TEXT
attempts	INTEGER
correct	INTEGER

The analytics table is used to record learner performance from previous quiz activities. This table serves as the basis for the application's adaptive learning feature.

During quiz activities, the system records the number of attempts and correct answers for each idiom encountered.

The database is updated through the `update_analytics()` function every time a user submits an answer to a multiple-choice question in Quiz mode. The system then automatically records the increased number of attempts and the corresponding number of correct responses. Afterwards, some learning statistics, which will be further explained in Section 4.2.5, can be calculated, such as:

  - Total practiced idioms

  - Mastered idioms
  - Weak idioms
  - Average accuracy rate

  The stored data allows the system to identify areas where users require additional practice and to generate recommendations that support a more personalized learning experience. These metrics help users monitor and improve their learning progress through feedback displayed in the Learning Analytics dashboard.

4.1.5. Streamlit

In this project, Streamlit is used as the graphical user interface and is deployed on Hugging Face Spaces. Through this platform, users do not need to perform any local installation to access the application, as it is directly available through a web browser. Hugging Face Spaces acts as the cloud hosting platform for the application.

The application was developed using various Streamlit components to provide an interactive and user-friendly interface. The main components used in the implementation are described below:

- **Application Layout:** The components `st.set_page_config()` and `st.title()` were used to configure the application window and display the application title.
- **Navigation Menus:** The component `st.sidebar.selectbox()` was used to allow users to switch between different application modes.
- **User Input Forms:** The components `st.text_input()`, `st.text_area()`, `st.selectbox()`, and `st.radio()` were used to collect user input. These components enable users to enter text for idiom detection, select topics and idioms, choose quiz answers, and search for idioms.
- **Buttons:** The component `st.button()` was used to trigger application functionalities such as adding favourites, removing favourites, practising idioms, detecting idioms, submitting answers, and generating new questions.
- **Displaying Results:** The components `st.write()`, `st.markdown()`, `st.subheader()`, `st.success()`, `st.error()`, `st.warning()`, and `st.info()` were used to display idiom meanings, literal translations, detection results, learning analytics, and quiz feedback.
- **Audio Playback:** The component `st.audio()` was used to play audio pronunciations generated using the Google Text-to-Speech (gTTS) library.
- **Learning Analytics Dashboard:** The components `st.metric()`, `st.progress()`, `st.bar_chart()`, and `st.dataframe()` were used to present user performance

statistics, weak idioms, mastered idioms, experience points (XP), and accuracy metrics in an interactive manner.

- **Session State Management:** The component `st.session_state` was used to preserve quiz progress and user data during page refreshes and navigation between application sections.
- **Expandable Sections:** The component `st.expander()` was used to display additional information about detected idioms while keeping the interface organized and uncluttered.
- **Performance Optimization:** The decorators `@st.cache_data` and `@st.cache_resource` were used to improve application performance by reducing repeated loading of machine learning models, datasets, and other computational resources.

4.2. Application Modes

4.2.1. Explore Idioms Mode

This mode is the first mode that appears on the main page of the application and applies the core concept of the project. It allows users to explore the list of idioms previously configured in the `idiom2.json` file. There is a search button where users can freely type any idiom that comes to mind. If they want to explore idioms by topic, there is also a topic filtering function that lists idioms by the selected topic.

Once an idiom is chosen, the system displays the corresponding natural and literal translations, example sentences (if available), and audio pronunciation for both languages.

For literal translation, the implementation requires the Hugging Face transformers library along with specific tokenization dependencies:

CÓDIGO 4.7. Installation of required NLP libraries.

```
pip install transformers==4.1.1 sentencepiece==0.1.94
pip install mosestokenizer==1.1.0
```

CÓDIGO 4.8. Importing MarianMT.

```
from transformers import MarianMTModel, MarianTokenizer
```

Then, initialize models that can translate English to Spanish.

The translation process begins by initializing the models designed for English-to-Spanish translation using the `Helsinki-NLP` weights.

CÓDIGO 4.9. Model and Tokenizer Initialization

```
model_name = "Helsinki-NLP/opus-mt-en-es"
tokenizer = MarianTokenizer.from_pretrained(model_name)
model = MarianMTModel.from_pretrained(model_name)
```

Subsequently, a helper function was developed to process a batch of text. This function manages the tokenization, the forward pass through the machine translation model, and the decoding of the resulting Spanish output.

CÓDIGO 4.10. Literal translation helper function

```
def translate_literal(text):
    # Ensure model and tokenizer are loaded
    tokenizer, model = load_translation_model()

    # Tokenize input text with truncation
    inputs = tokenizer(text, return_tensors="pt", truncation=True)

    # Generate translation with a maximum of 100 new tokens
    translated = model.generate(**inputs, max_new_tokens=100)

    # Decode and return the final string
    return tokenizer.decode(translated[0], skip_special_tokens=True)
```

Based on these findings, it can be concluded that MarianMT indeed an offline alternative to Google Translate which is available for direct translation and also back-translation [13].

Since this project uses Python 3.14.3, which exceeds the minimum required version of Python 3.7, gTTS can be installed and executed without compatibility issues using the command ‘pip install gTTS’.

CÓDIGO 4.11. Implementation of audio generation using gTTS.

```
@st.cache_data
def generate_audio(text, lang="en"):
    safe_text = text.replace(" ", "_")
    file_path = CACHE_DIR / f"{safe_text}.mp3"

    if not file_path.exists():
        tts = gTTS(text=text, lang=lang)
        tts.save(file_path)

    return str(file_path)
```

As shown in python code 4.11, the function defined to support audio pronunciation. Each selected idiom is associated with a "Play" button. When the button is pressed, the system automatically generates an audio version of the expression and makes it available for playback within the user interface.

A small "Add to Favorites" button, represented by a star icon, is located below the audio section, allowing users to save their favourite idioms for future reference.

Figure 4.1 shows an example where the idiom "back to square one" was selected by a user.

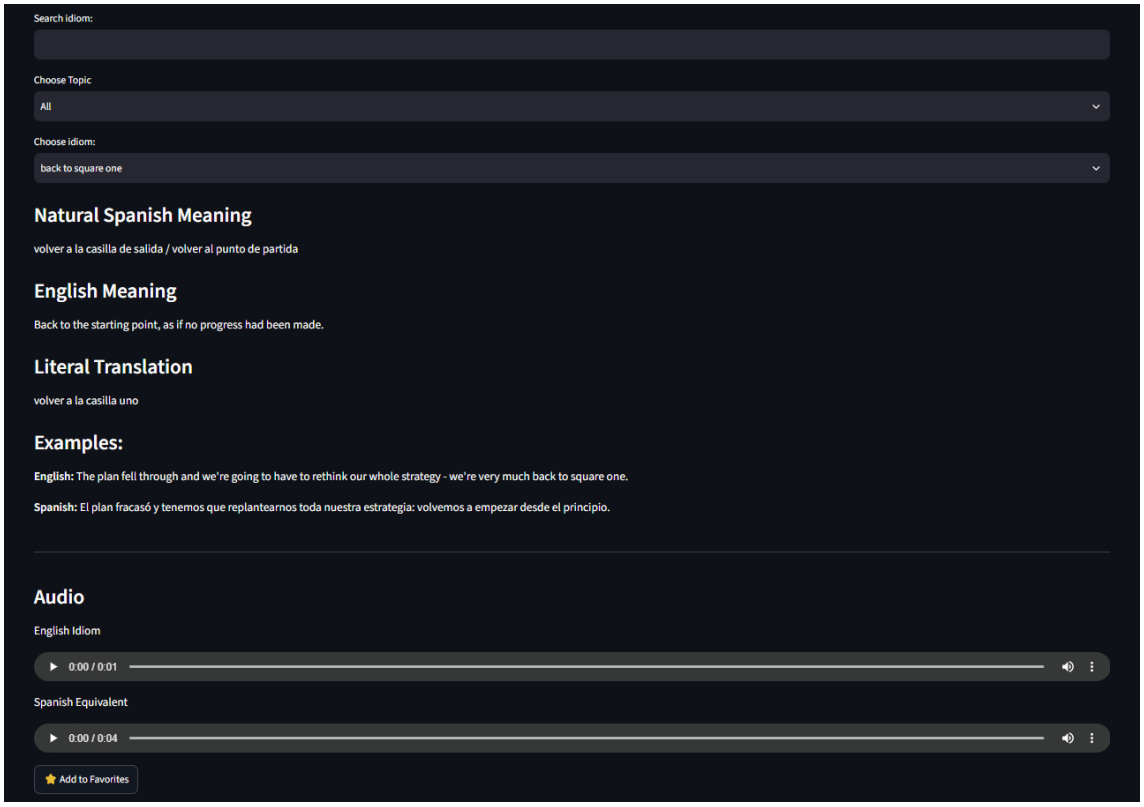


Fig. 4.1. Explore Idioms Mode

4.2.2. Idioms in Sentences Mode

Idioms in Sentences mode is a mode that helps learners identify idioms used in their sentences. The detected idiom will then be highlighted so that users can clearly see the idiom, its translation, corresponding example sentences, and audio pronunciations. Moreover, users can write sentences or paragraphs without an idiom, and the system will suggest available idioms that suit their sentences using Artificial Intelligence (AI). For this purpose, `paraphrase-multilingual-MiniLM-L12-v2` was configured.

The system uses both traditional and modern Natural Language Processing (NLP) approaches. Traditional NLP approaches rely on rule-based detection techniques that identify keywords or predefined patterns within a text. These rules are followed explicitly by the system to ascertain the presence of an idiom. This deterministic approach depends on developers' predefined logical conditions or thresholds, which are manually created using domain knowledge or application requirements [74]. Rule-based detection compares and detects exact sentences or words that appear in the `idiom2.json` file.

However, this approach has several limitations. Some idioms consist of Multiword Expressions (MWEs), which are described as idiomatic or peculiar phrases that transcend word boundaries and make up a significant percentage of human lexicons (estimated at over 41 percent in WordNet 1.7). Therefore, it is prone to flexibility problems, where these words-with-spaces often fail to account for syntactic variation. For example, a user may mistakenly write "bite a bullet" instead of "bite the bullet", as defined in the predefined idiom list. Another issue is lexical proliferation, where listing every inflected form or variant is inefficient, such as "wet/wetted/wets". As a result, the system may fail to detect the intended idiom [75].

CÓDIGO 4.12. Rule-based idiom detection function.

```
def detect_idioms(text, idioms):
    tokens = list(re.finditer(r"\b\w+\b", text))
    text_words = [t.group().lower() for t in tokens]
    text_lemmas = [lemmatize_word(w) for w in text_words]

    found = []
    spans = {}

    for idiom in idioms:
        idiom_words = idiom.lower().split()
        idiom_lemmas = [lemmatize_word(w) for w in idiom_words]
        n = len(idiom_lemmas)

        for i in range(len(text_lemmas) - n + 1):
            if text_lemmas[i:i+n] == idiom_lemmas:
                found.append(idiom)
                start = tokens[i].start()
                end = tokens[i + n - 1].end()
                spans[idiom] = text[start:end]
                break

    return found, spans
```

Listing 4.12 presents the rule-based idiom detection function used in Idioms in Sentences Mode. In order to normalize distinct forms of the exact same word, the input text is first tokenized and lemmatized. Each idiom in the dataset is made into its base form and checked against groups of words that come one after another in the text. If a match is found, the idiom is recognized and the part of the text that matches is recorded. This approach allows idioms to be detected despite variations in grammatical form, such as changes in tense and inflection . For example, "break a leg" and "broke a leg" both are recognized as the idiom "break a leg".

Using this method, the system can effectively employ basic string-matching techniques to identify predefined idiomatic expressions.

AI-based anomaly detection uses machine learning models to detect outliers and sub-

sequently learn patterns from data. This method helps solve more complex problems due to its ability to adapt to data patterns using techniques such as autoencoders, clustering, or time-series forecasting to identify trends that humans might overlook [74]. The model is able to identify the underlying meaning of text. It examines the semantic meaning and sentential context using pre-trained language models rather than relying entirely on exact keyword matching [76].

For example, a rule-based system may only detect:

"spill the beans"

when it appears in its exact, canonical form. In contrast, semantic models can infer the intended meaning of a sentence even when no predefined pattern is present. For example:

"He revealed the secret unexpectedly."

carries a semantic interpretation comparable to the expression "spill the beans", even though there is no precise lexical overlap.



Fig. 4.2. Flowchart AI-detection

The flowchart of the semantic AI model used to recommend an appropriate idiom for the user's content is displayed in Figure 4.2. The user input is transformed into a sentence embedding using the Multilingual MiniLM model, which then uses semantic similarity to compare the embeddings of the stored idiom meanings. If the score reaches the predetermined threshold, the idiom with the highest similarity score is chosen.

CÓDIGO 4.13. Semantic idiom detection function using SentenceTransformers.

```
def detect_idioms_ai(text, idiom_map, threshold=0.5):
    model = load_similarity_model()

    idiom_map_tuple = tuple(
        (idiom, build_embedding_text(idiom, info))
        for idiom, info in idiom_map.items()
    )

    idioms, meaning_embeddings = _get_idiom_meaning_embeddings(
        idiom_map_tuple
    )

    text_embedding = model.encode(text, convert_to_tensor=True)
    scores = util.cos_sim(text_embedding, meaning_embeddings)[0]

    best_idx = int(scores.argmax())
    best_score = float(scores[best_idx])

```

```
    if best_score < threshold:
        return []

    return [idioms[best_idx]]
```

Based on Listing 4.13, the SentenceTransformer model is used to first transform a sentence entered by the user into an embedding. Next, cosine similarity is used to compare the generated embedding with the idiom meanings' embeddings. After determining which idiom has the highest similarity score, the system compares it to a predetermined threshold of 0.5. The related idiom is recommended as the detected idiom if the score reaches or surpasses the threshold, if not, no idiom is returned. The semantic approach is designed to identify idioms based on the semantic similarity between the user's text and the stored idiom meanings, rather than relying solely on exact string matching.

The purposed application combines both detection approaches. While AI semantic detection improves the system's capacity to comprehend contextual meaning and manage more complicated language usage scenarios, rule-based detection offers quick and accurate identification of recognized idioms.

4.2.3. Quiz Mode

This mode allows users to challenge themselves by practicing quiz questions. The quiz generation is configured using two different methods: question generation from example sentence datasets and AI-generated questions. There are four multiple-choice answers consisting of four different idioms from which the user can choose. Each time an answer is selected, the system indicates whether it is correct or incorrect and displays the correct answer if the user's response is wrong. Users can then navigate to a new question by pressing the "New Question" button.

FLAN-T5 is used for AI-generated quiz creation. As an instruction-tuned transformer model, FLAN-T5 is capable of generating contextually relevant text from natural language prompts. In the application, it is utilized to generate quiz questions based on English idioms, enhancing the variety and adaptability of learning activities.

CÓDIGO 4.14. Python function for normalizing sentence structure prior to matching.

```
def normalize_structure(sentence):
        """
    Simplify sentence structure for repetition detection.
    """

    sentence = sentence.lower()

    # remove punctuation
    sentence = re.sub(r"[^\w\s]", "", sentence)

```

```
    # remove common subjects
    starters = [
        "i", "he", "she", "they", "we", "my boss",
        "the company", "someone", "people", "our team"
    ]

    words = sentence.split()

    if len(words) >= 2:
        first_two = " ".join(words[:2])

        if first_two in starters:
            return " ".join(words[2:5])

    return " ".join(words[:3])
```

Listing 4.14 shows the normalize_structure function used to reduce repetition among AI-generated quiz questions.The function standardizes the input by converting the sentence to lowercase, removing punctuation marks, filtering out common subject starters including pronouns and generic nouns, and extracting a representative root word sequence in order to compare patterns.

CÓDIGO 4.15. Prompt template for generating paraphrased sentences with FLAN-T5.

```
prompt = f"""Paraphrase the following sentence.
Rules:
- Keep the idiom "{idiom}" exactly unchanged.
- Keep the same meaning.
- Return only one sentence.
- Do not explain anything.
Sentence:{base_sentence}"""
```

CÓDIGO 4.16. Text generation generation parameters for model inference.

```
inputs = tokenizer(prompt, return_tensors="pt", truncation=True)
outputs = model.generate(
    **inputs,
    max_new_tokens=50,
    do_sample=True,
    temperature=0.7,
    top_p=0.9,
    repetition_penalty=1.2,
)
```

The AI-generated sentence is produced using the FLAN-T5 model, as shown in Listing 4.16. Prior to being sent to the model, the prompt is tokenized and transformed into tensors. The model then uses controlled sampling parameters, such as a maximum output length of 50 tokens, a temperature value of 0.7, a top-p value of 0.9, and a repetition

penalty of 1.2, to produce a paraphrased sentence.

CÓDIGO 4.17. Post-generation validation and filtering criteria for candidate sentences.

```
if not sentence or len(sentence.split()) < 8:
    continue
if len(sentence.split()) < len(base_sentence.split()) * 0.6:
    continue
if not idiom_pattern.search(sentence):
    continue
if sentence.strip().lower() == base_sentence.strip().lower():
    continue
if any(leak in sentence.lower() for leak in leakage_phrases):
    continue
```

Listing 4.17 demonstrates that the generated sentence should satisfy several criteria. The minimum sentence length must be at least eight words, the sentence must have sufficient length relative to the original sentence, the target idiom must be present, the generated sentence must differ from the original sentence, and any unwanted prompt-related text must be excluded.

This mode also comprises two other main stages: distractor generation and adaptive quiz logic.

Distractor Generation

Distractor generation is a function that creates incorrect answer options displayed alongside the correct answer. Users are encouraged to learn the idiom by understanding its meaning rather than relying on random guessing. This approach also increases the difficulty of the question. The system randomly selects three distinct idioms from the idiom repository after choosing the target idiom. The correct answer is then combined with these distractors and shuffled before being displayed to the user.

Listing 4.18 presents the distractor generation function, which begins by removing the correct idiom from the candidate pool. It then randomly selects three distractors, combines them with the correct answer, and shuffles the options to provide unpredictable answer positions.

CÓDIGO 4.18. Distractor generation for multiple-choice questions.

```
def generate_distractors(correct_idiom, idiom_map, n=3):
    correct_topic = idiom_map[correct_idiom].get("topic", "General")

    same_topic_pool = [
        i for i in idiom_map
        if i != correct_idiom and idiom_map[i].get("topic", "General
            ") == correct_topic
    ]
```

```
    other_pool = [
        i for i in idiom_map
        if i != correct_idiom and idiom_map[i].get("topic", "General
            ") != correct_topic
    ]

    random.shuffle(same_topic_pool)
    random.shuffle(other_pool)

    distractors = same_topic_pool[:n]
    if len(distractors) < n:
        distractors += other_pool[: n - len(distractors)]

    options = distractors + [correct_idiom]
    random.shuffle(options)
    return options
```

Adaptive Quiz Logic

Adaptive quiz selection is incorporated into the system to improve user performance by adapting quiz content according to the user's previous performance. The objective is to provide additional practice with an appropriate degree of difficulty. The steps of the mechanism are:

- Quiz results are stored in SQLite.
- The number of attempts and correct answers is recorded.
- Accuracy is calculated (accuracy = correct answers / number of attempts).
- Weak idioms are identified (accuracy is less than 60% with at least two attempts).
- New quizzes are generated by prioritizing weak idioms.

After a quiz question is answered, the adaptation process is initiated by updating the analytics table, which increases the number of attempts for the corresponding idiom. The number of correct responses is incremented when the answer is correct. Next, these values are persistently retained in SQLite, ensuring that learning progress is preserved across sessions.

Equation 4.1 below shows how the accuracy score is computed for each idiom by the system after updating the statistics. It is important to note that the data must be sufficient before drawing any conclusions, meaning that each specific idiom must be attempted at least twice. Moreover, if the accuracy falls below 60%, the idiom is classified as a weak idiom and requires further practice.

$$Accuracy = \frac{Correct\ Answers}{Total\ Attempts} \tag{4.1}$$

The application uses the `get_weak_idioms()` function to retrieve all weak idioms from the database before recreating the quiz. If there are weak idioms, the system limits the pool of candidates to those idioms and chooses one at random for the subsequent question. The system chooses from the entire idiom dataset if no weak idioms are identified. This approach permits coverage of new information while guaranteeing that students are exposed to phrases they previously answered incorrectly.

Additionally, previously used questions will not be repeated, as the application already maintains a record of them through the `used_questions` and `used_structures` session variables. As a result, recently introduced idioms and sentence structures are temporarily removed from the selection process, promoting more diversity in the quizzes that are produced.

SQLite is used to store the analytics table, which provides the information required for this adaptive mechanism. The table records the number of attempts and correct answers for each idiom, helping to estimate learner proficiency and support personalized future quiz questions.

4.2.4. Favorites Mode

From the previously discussed Explore Idioms mode, there is a small button at the bottom labelled "Add to Favorites", which can be selected by the user. Favorites mode is a mode where users are able to store idioms that they find difficult, useful, or interesting for future reference. They can also remove an idiom from their favorites list if they are no longer interested in it.

The data from this mode is stored by the system in SQLite, which automatically loads previously saved favorites to ensure persistence across user sessions. Hence, the saved idioms remain available even after the application is closed. Listing 4.19 below shows the code used to record a saved idiom together with its corresponding timestamp, indicating when the idiom was added. The idiom field in the table is designated as the primary key to prevent duplicate entries from being stored.

CÓDIGO 4.19. Creation of the favorite table in SQLite.

```
CREATE TABLE IF NOT EXISTS favorite (
    idiom TEXT PRIMARY KEY,
    timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

By enabling users to compile their own list of idiomatic expressions, the Favorites mode supports personalized learning. Without browsing the entire repository, learners

can easily review their saved idioms at any time. This function facilitates self-directed learning and provides quick access to expressions that require additional practice or review.

4.2.5. Learning Analytics Mode

The Learning Analytics Dashboard displays user performance in a more interactive way. At the top of the page is the XP progress indicator, which shows the number of XP collected by the user so far. This system introduces an element of gamification, encouraging learners to continue practising and improving their performance. For each correct answer, the system awards 5 XP, as shown below.

CÓDIGO 4.20. Updating the user's XP score.

```
st.session_state.xp += 5
```

Then, the user's progress is also illustrated by four different metrics:

- Mastered. Idioms answered correctly at least 80% of the time

CÓDIGO 4.21. Mastered idiom calculation.

```
mastered = len([
    i for i,a,c,acc in stats
    if acc >= 0.8 and a >= 2
])
```

- Practiced. Idioms you have attempted at least once.

CÓDIGO 4.22. Practiced idiom calculation.

```
practiced = len([
    i for i,a,c,acc in stats
    if a > 0
])
```

- Weak. Idioms with accuracy below 60%.

CÓDIGO 4.23. Weak idiom calculation.

```
weak = len([
    i for i,a,c,acc in stats
    if acc < 0.6 and a > 0
])
```

- Accuracy. Your average correct rate across all attempted idioms.

CÓDIGO 4.24. Average accuracy calculation.

```
avg_acc = np.mean([acc for _,a,_,acc in stats if a > 0]) if stats
    else 0
```

In addition, the analytics section is followed by a performance chart showing all idioms based on user performance. Hence, users will be able to compare their performance across distinct idiomatic expressions and quickly recognize weaker areas.

Then, the "Idioms You Should Practice" section displays weak idioms that should be practiced by the user in future quizzes. This feature helps students focus on expressions that need more reinforcement.

Last but not least, detailed statistics are presented, showing multiple idiom entries together with their corresponding Attempts, Correct, and Accuracy columns.

Adaptive learning is greatly supported by the analytics database. The system can identify idioms with low accuracy rates and prioritize them in subsequent quiz sessions by tracking user performance over time. As a result, learners are provided with additional practice opportunities in areas where improvement is needed, creating a more personalized learning experience.

4.3. Graphical User Interface

4.3.1. Front-End Design

This section presents the graphical user interface (GUI) of the application, showcasing each of the five learning modes: Explore Idioms, Idioms in Sentences, Quiz Time, Favorites, and Learning Analytics. Each mode was designed to support idiom learning through a distinct feature, offering an interactive approach to learning and using idioms across English and Spanish.

Figure 4.3 illustrates the five different modes available from the sidebar of the Streamlit interface.

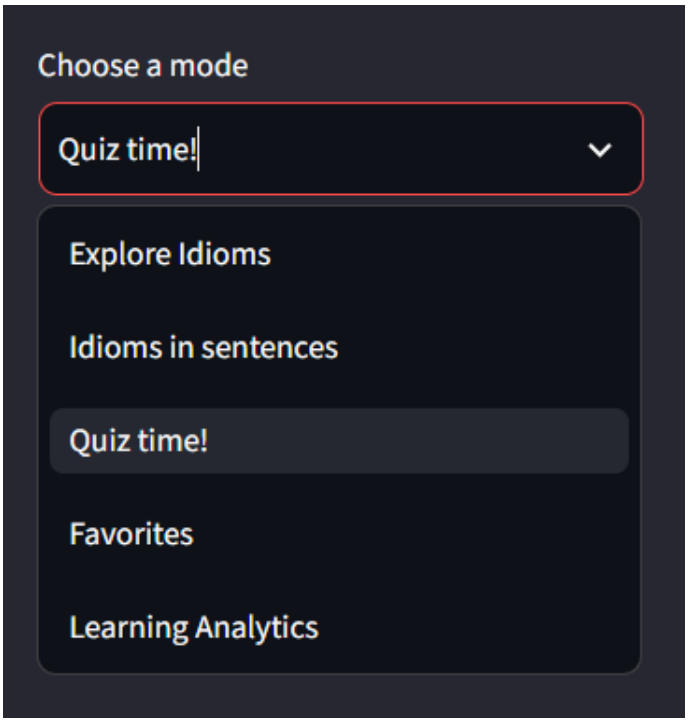


Fig. 4.3. Main modes of the application

Explore

In this mode, several functions are displayed, as shown in Figure 4.4, where users are able to choose any idiom from 32 unique topics.

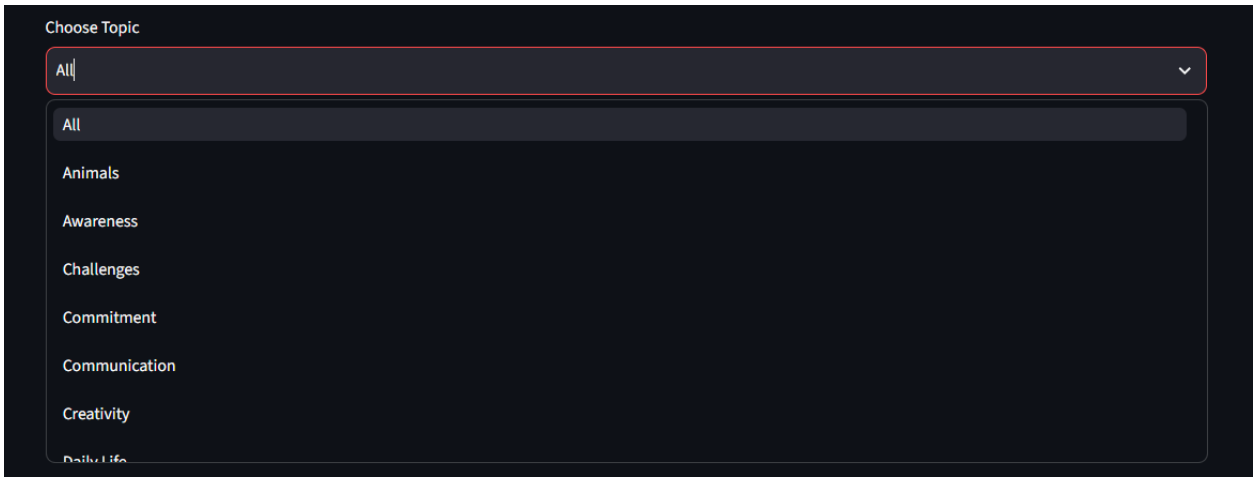


Fig. 4.4. Filter by Topic

For every idiom selected, the system displays the Natural Spanish Meaning, which represents the translation commonly used by native speakers. Meanwhile, the Literal Translation shows the direct word-for-word translation, allowing users to differentiate between the two types of translation.

In addition to that, available example sentences are also displayed, as shown in Figure 4.5.

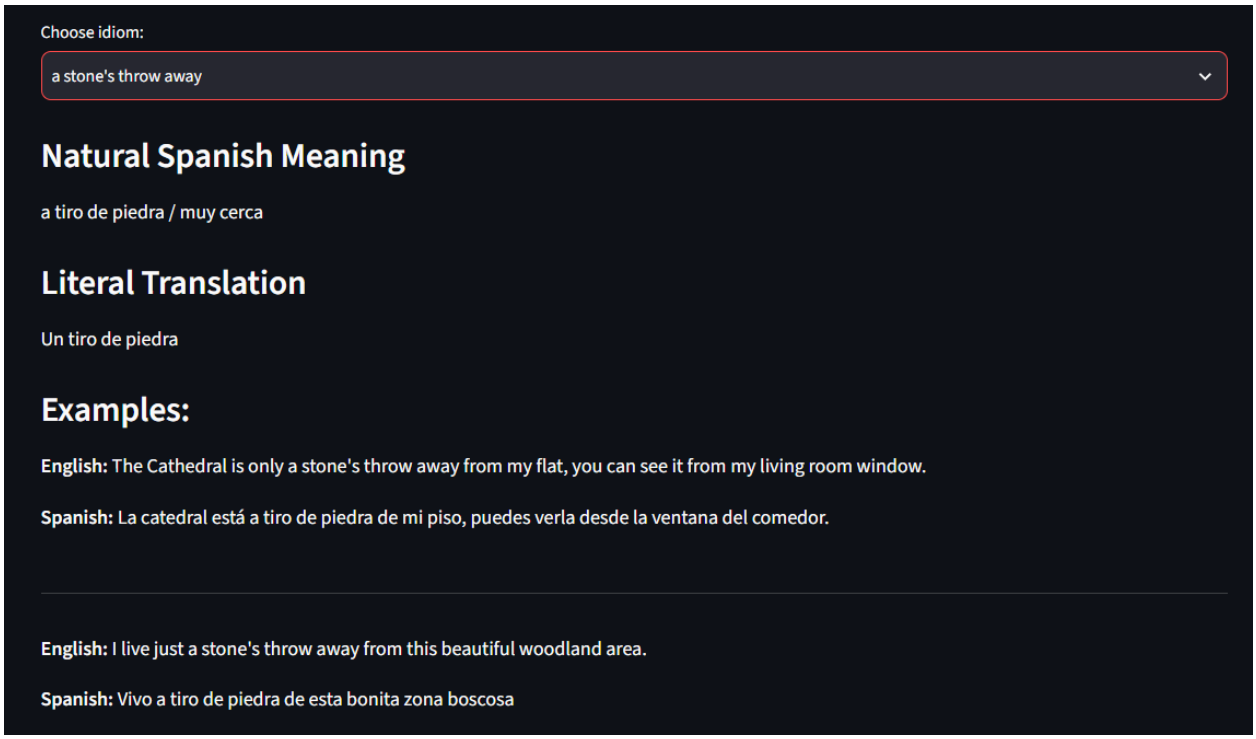


Fig. 4.5. Idiom translation

Figure 4.6 presents the audio pronunciation feature of the selected idiom.



Fig. 4.6. Audio pronunciation

Idioms in Sentences

This mode allows users to enter free text to check whether an idiom is present in the sentence, as shown in Figure 4.7. The detected idiom is highlighted along with its corresponding meaning, examples of sentences, and audio pronunciation. Some users may not realize that a common expression they use is actually an idiom.



Fig. 4.7. Idiom detection

In addition, the application can suggest an idiom that fits the text even when no idiom is present, as shown in Figure 4.8 and Figure 4.9. Hence, users can learn and apply the suggested idiom in their daily conversations.

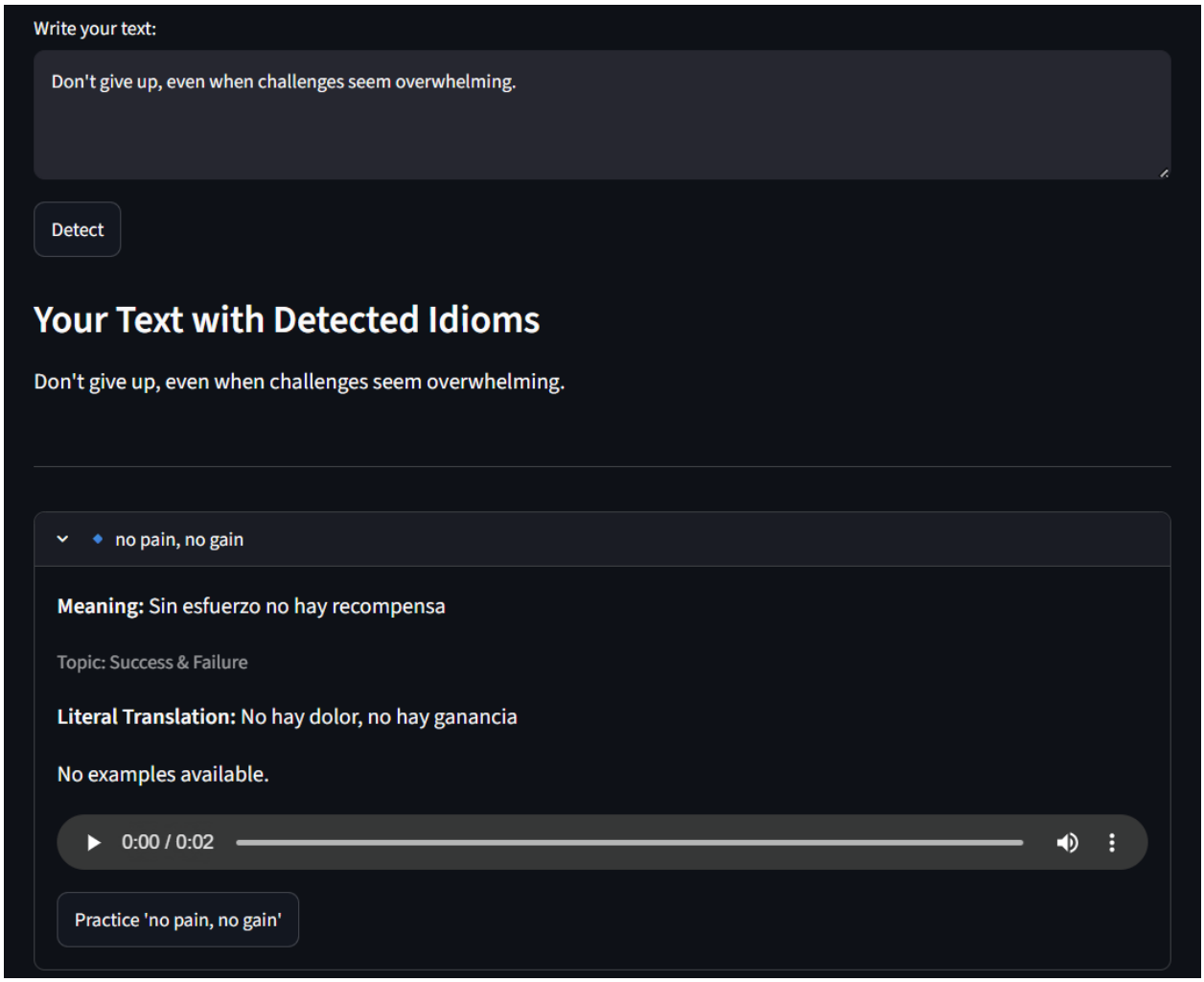


Fig. 4.8. Idiom suggestion

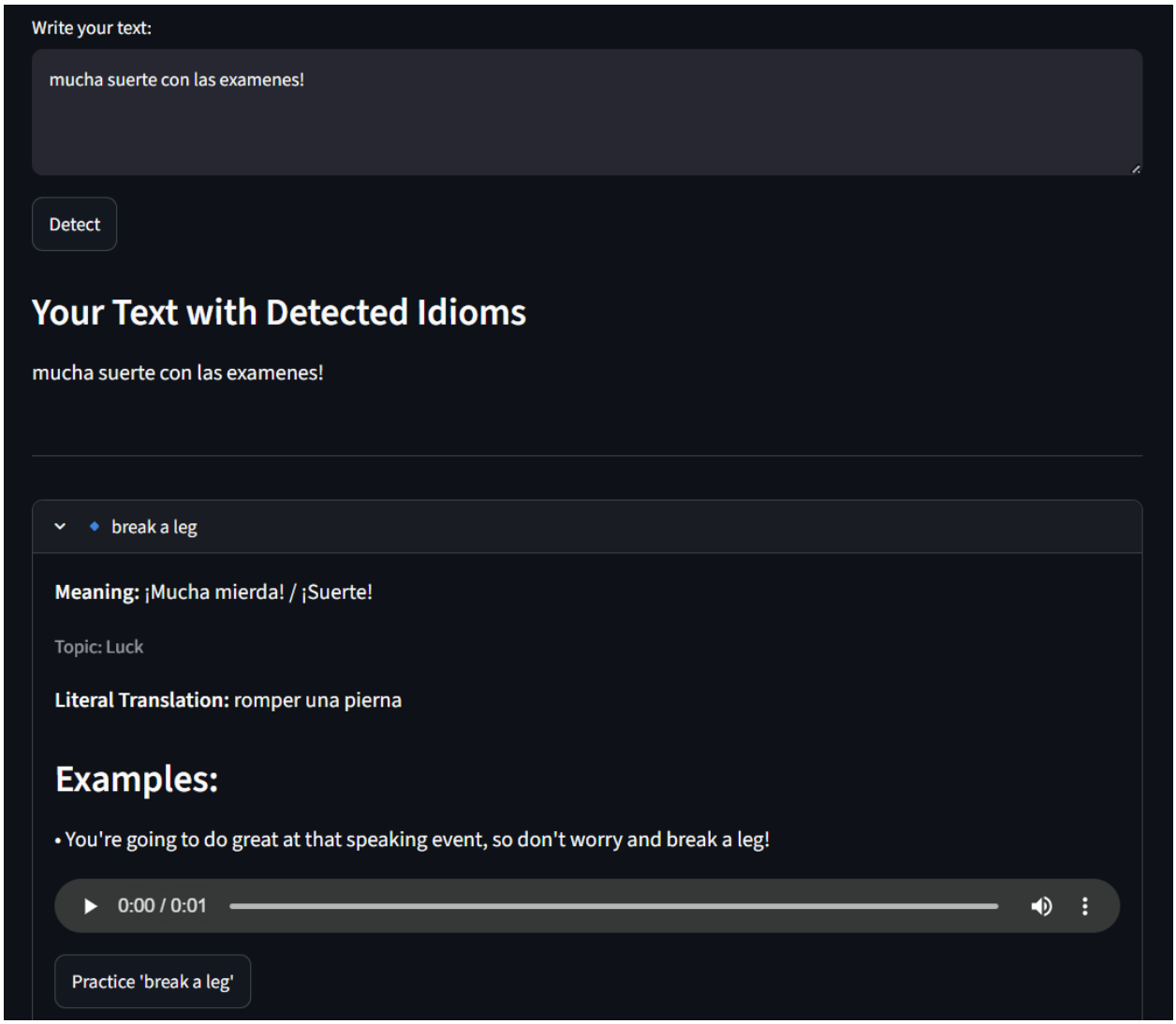


Fig. 4.9. Idiom suggestion 2

Quiz time!

This section is where users can challenge themselves with multiple-choice questions. Every correct answer adds +5 XP, as shown in Figure 4.10, while any incorrect answer also displays the correct answer, as shown in Figure 4.11. There is also a button that allows users to navigate to the next question.

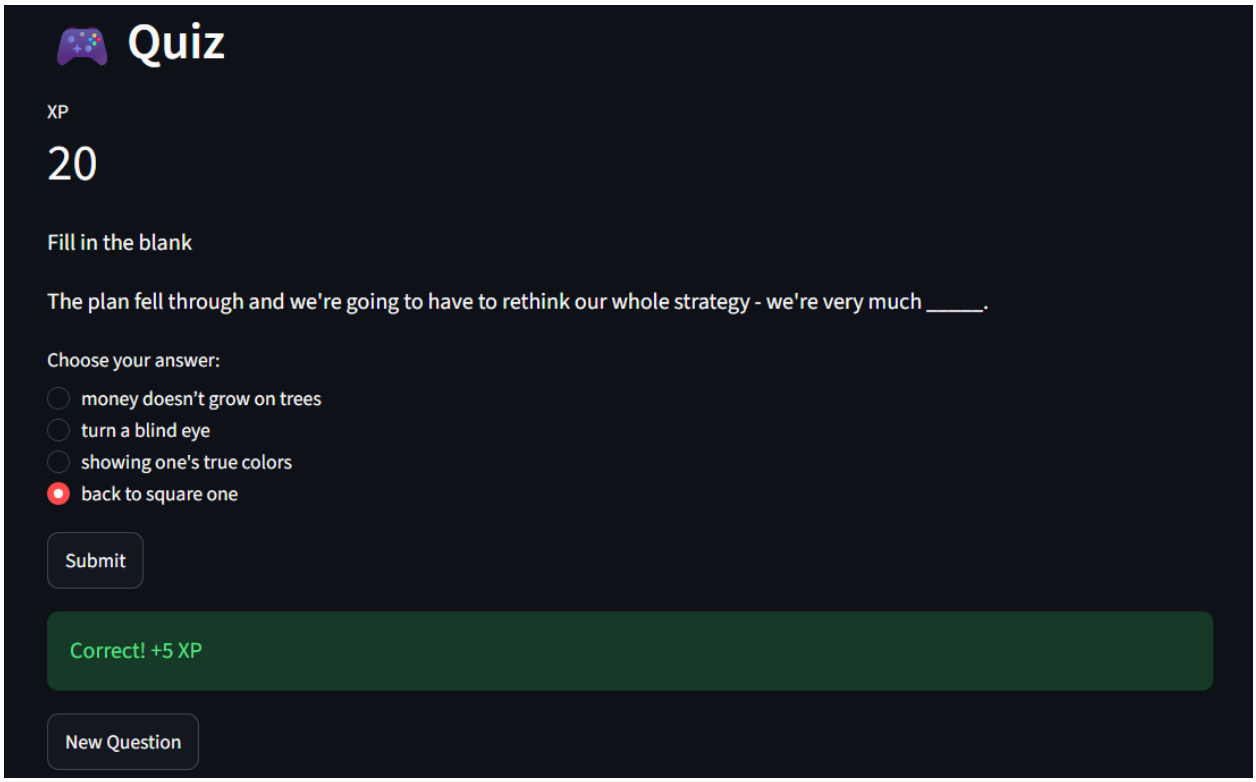


Fig. 4.10. Example of Correct Idiom Selection

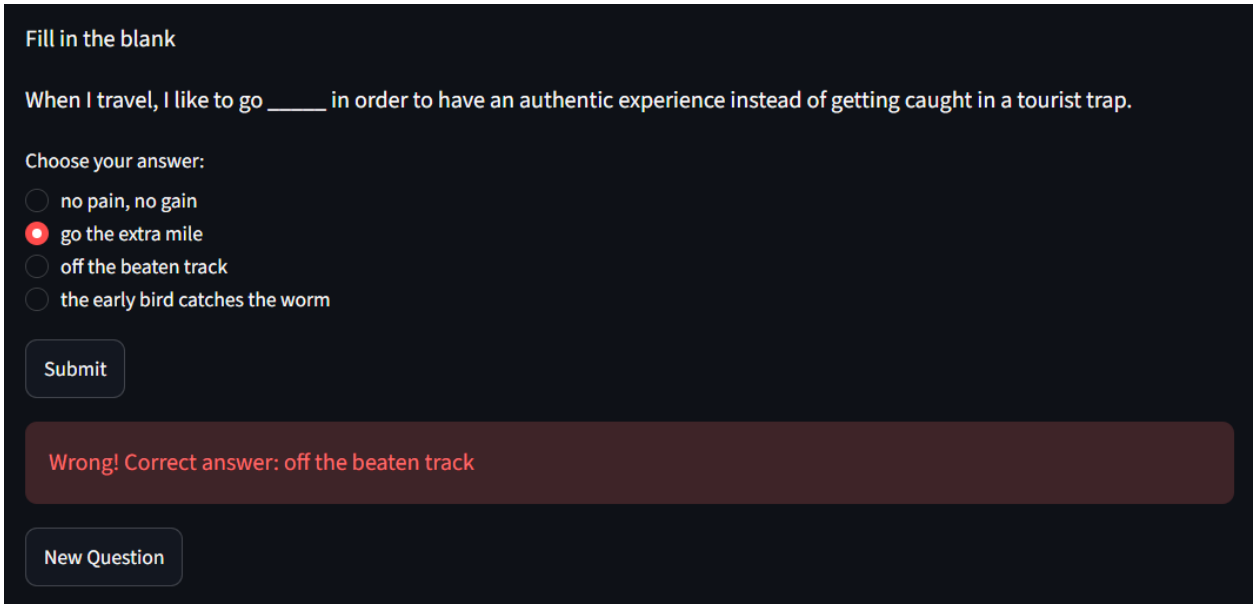


Fig. 4.11. Example of Wrong Idiom Selection

Analytics

The Learning Analytics mode illustrates a dashboard used to track user progress.

Figure 4.12 presents the number of XP accumulated by the user together with four different metrics, as discussed in Section 4.2.5.

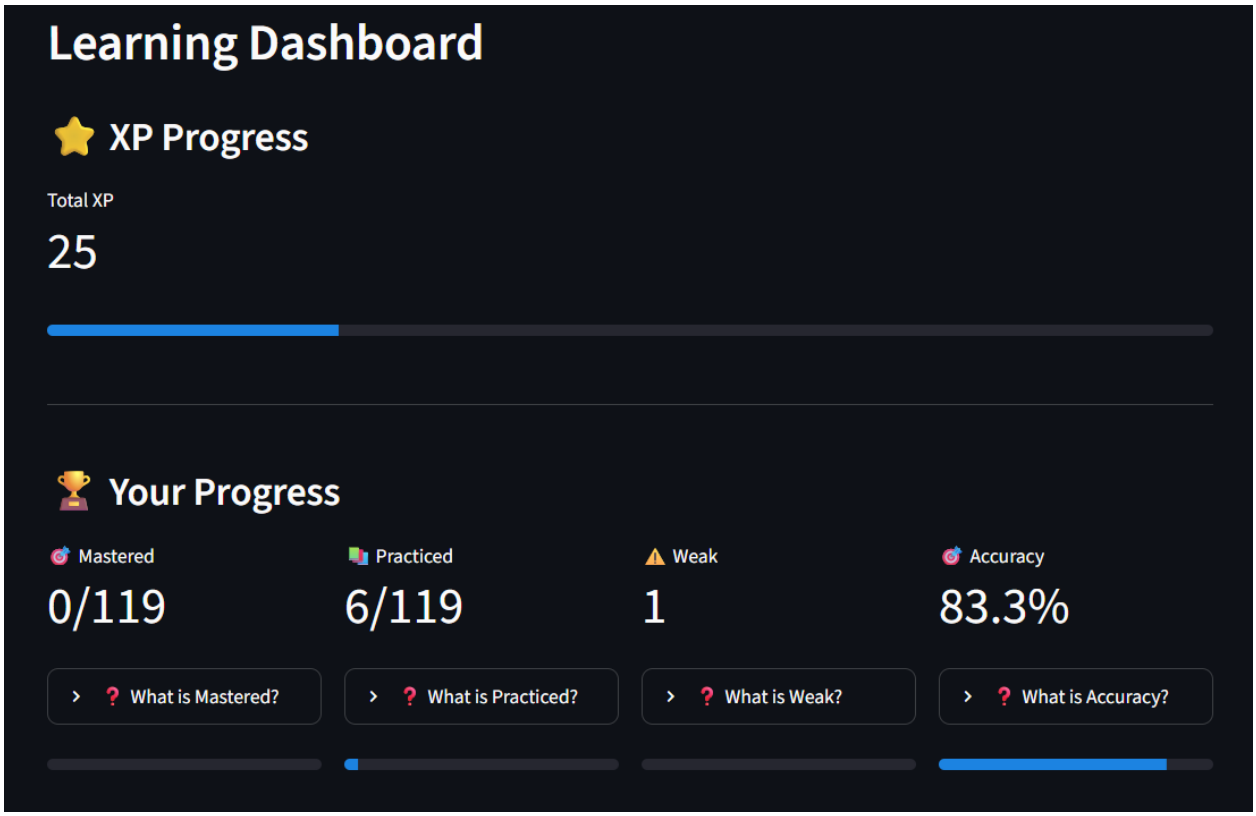


Fig. 4.12. Dashboard of Analytics

Figure 4.13 displays user performance through a chart consisting of idioms previously practised in the Quiz mode. There is also a section showing Idioms to Practice, which contains idioms that were answered incorrectly, as well as detailed statistics for reference.

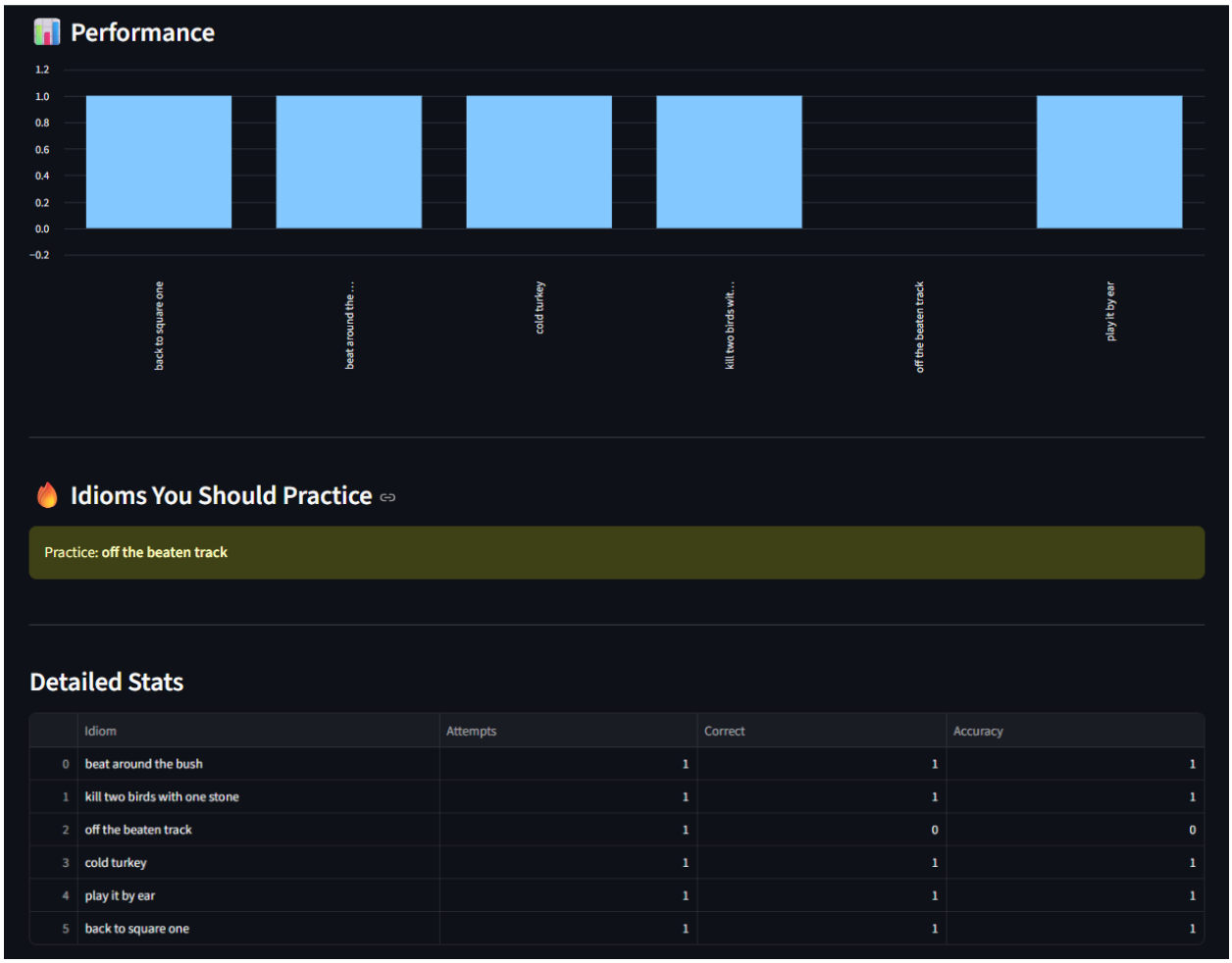


Fig. 4.13. Chart and Detailed Stats

Favorites

From the mode discussed in Part 4.3.1, there is a small button with a star icon that allows users to add the selected idiom to Favorites, as shown in Figure 4.14.



Fig. 4.14. Addition to Favorites

Figure 4.15 illustrates the Favorites interface. This function enables users to view, organize, and delete their saved idioms. Preferred idioms are stored in a SQLite database, which facilitates ongoing storage during different application sessions. This functionality assists users in reviewing significant idiom without needing to browse through the complete idiom collection.

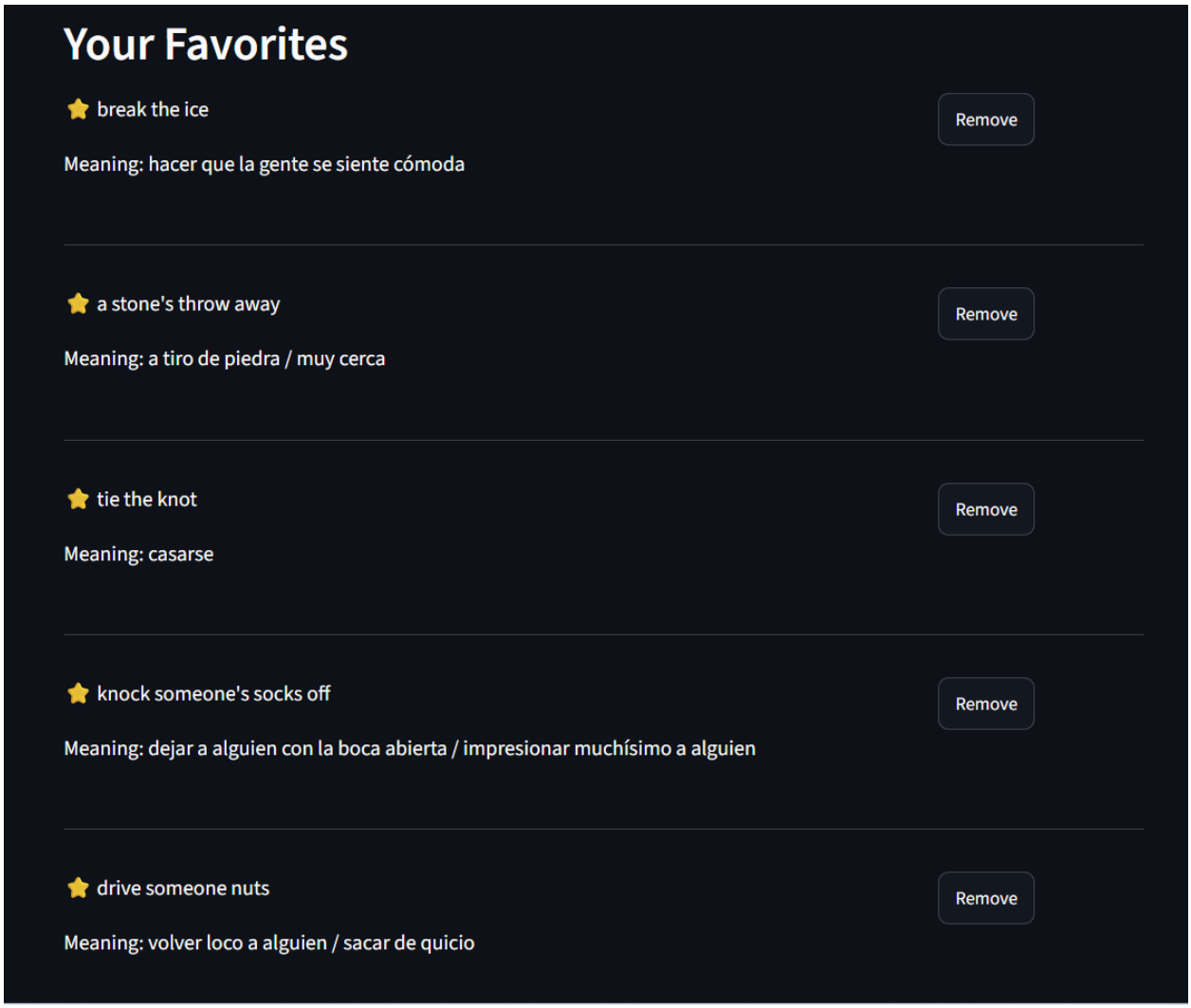


Fig. 4.15. Saved Idioms

5. RESULTS AND EVALUATION

5.1. Functional Testing

TABLE 5.1. DETAILED FUNCTIONAL TESTING RESULTS

Feature	**Expected Outcome**	**Result**
Literal Translation	Spanish translation displayed for selected idiom	Success
Audio Generation	Pronunciation audio generated and played correctly	Success
Idiom Detection	Idioms detected and highlighted in user text	Success
Quiz Generation	Fill-in-the-blank quiz generated successfully	Success
Favorites Management	Idioms added and removed from favorites list	Success
Learning Analytics	Statistics and charts displayed correctly	Success
Adaptive Learning	Weak idioms prioritized in future quizzes	Success

Table 5.1 summarizes the functional tests conducted on the application and the corresponding results. First, the literal translation, natural Spanish translation, and corresponding idiomatic expressions are displayed correctly as expected. The audio generation module is also able to produce English pronunciation audio files using gTTS successfully.

The idiom detection feature on the next page of the application is able to successfully highlight idiomatic expressions present within the user-input text.

The next function evaluated was the Quiz section, where fill-in-the-blank questions with multiple-choice answers are generated successfully. The Favorites feature, which allows users to add and remove desired idioms, is correctly configured and stored in the SQLite database. Finally, the Learning Analytics and adaptive learning components accurately monitor user performance and personalize upcoming quiz questions according to the weak idioms identified.

5.2. AI Evaluation

This section evaluates the three AI integration models involved in this project.

MarianMT

MarianMT was successfully validated by translating English idioms into their equivalent Spanish literal translations. The generated translations were used only for literal interpretation rather than semantic equivalence because idiomatic expressions often carry

figurative meanings. This aligns with the objective of the project, which is to generate grammatically correct literal Spanish translations for English idioms alongside their natural idiomatic meanings, thereby helping users better differentiate between the two meanings.

For the evaluation, two metrics have been utilized to conduct this evaluation, which are literal meaning preservation and grammatical correctness. The literal meaning preservation metric evaluates how MarianMT preserves the literal meaning word-by-word of the English idiom when translated into Spanish. Therefore, each MarianMT output is manually evaluated as correct if the Spanish literal translation generated accurately conveys the literal meaning of the original English expression. Otherwise, it is classified as incorrect. The accuracy of this metric is evaluated as shown below:

$$\text{Accuracy} = \frac{\text{Correct translations}}{\text{Total idioms}}$$

Meanwhile, the second metric evaluates the grammatical correctness of the Spanish translation, also as either correct or incorrect. The generated sentences are classified as grammatically correct by examining components such as the arrangement of words, verb forms, the correct use of articles, consistency in gender and number, as well as other fundamental grammar rules in Spanish. Consequently, a result might maintain the exact meaning but still contain grammatical mistakes. The formula used is shown below:

$$\text{Grammar Accuracy} = \frac{\text{Grammatically correct outputs}}{\text{Total outputs}}$$

This evaluation uses 50 idioms, chosen from different topics and difficulty levels from the application's dataset (`idiom2.json`) to provide a clear sample for manual assessment, enabling the evaluation of MarianMT's performance across a diverse array of idiomatic expressions. The 50 idioms were selected from the 119 idioms to ensure data quality, with the most significant and interesting idioms selected from diverse topics and various levels of difficulty. The evaluation was manually conducted by the author initially, and subsequently reviewed and verified by a native Spanish speaker to ensure the accuracy of the assessment.

Table 5.2 presents the results of the 50-idiom evaluation to study the performance of MarianMT. Literal meaning was successfully preserved in about 90% of the total idioms evaluated, while 86% were considered grammatically correct.

TABLE 5.2. EVALUATION OF GENERATED IDIOM TRANSLATIONS.

Metric	**Correct**	**Total**	**Accuracy**
Literal meaning preserved	45	50	90%
Grammatically correct	43	50	86%

Table 5.3 lists some results of the application in which the model correctly translates English idioms into literal Spanish translations. The complete results are available in the evaluation dataset[8]. Since the model was not designed specifically for idiomatic interpretation, it is expected that the generated output does not always reflect the true figurative meaning of the idiom.

TABLE 5.3. TRANSLATION QUALITY AND EQUIVALENCE EVALUATION.

English Idiom	**MarianMT Literal**	**Natural Spanish Equivalent**	**Literal Meaning**	**Grammar**
bite the bullet	muerde la bala	armarse de valor	✓	✓
hit the nail on head	golpear el clavo en la cabeza	dar en el clavo	✓	✓
kick the bucket	patear el cubo	estirar la pata	✓	✓
under the weather	bajo el tiempo	estar indispuesto	✓	X

FLAN-T5 base

Quiz questions are generated using the FLAN-T5 Base model, as illustrated in the following figures.

[8] https://github.com/kawaa099-dot/english-spanish-idioms-app/blob/main/evaluations/marianmt_test.csv

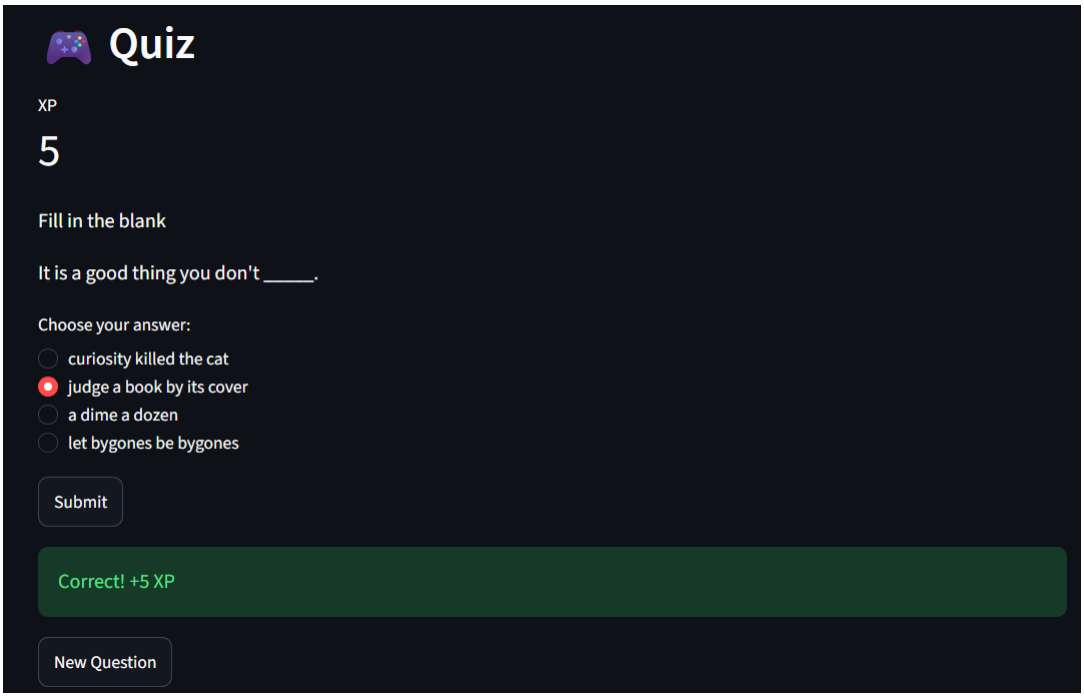


Fig. 5.1. AI-generated question 1

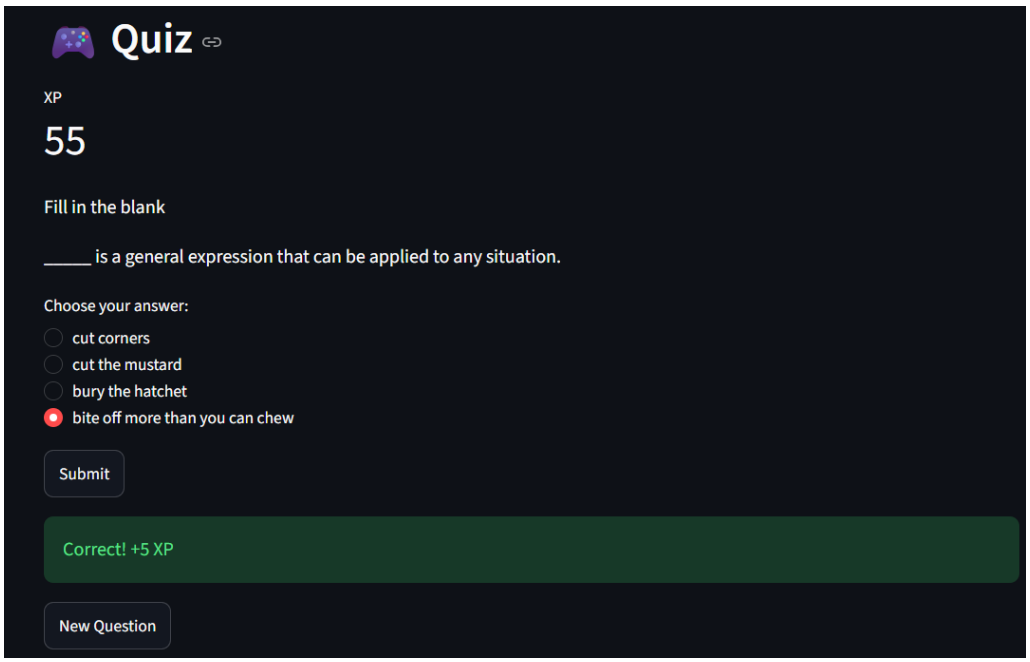


Fig. 5.2. AI- generated question 2

For the quiz evaluation, 100 sentences were generated[9], where the generation was successfully completed using the application itself without taking into account the slow AI generation between, where the user needs to press the "New Question" button again. From the 100 sentences, 93 sentences were considered valid questions, with no semantically

[9]The evaluation dataset is available at https://github.com/kawaa099-dot/english-spanish-idioms-app/blob/main/evaluations/quiz_evaluation_100.csv

broken questions where the idiom itself does not fit correctly into the blank's meaning, regardless of sentence quality. Then 95% of the generated sentences were grammatically correct with no broken sentences. Furthermore, 12 items were flagged as unmeaningful sentences, where the sentence itself does not make any sense. Thus, 88% were usable sentences, considering only meaningful items, and can be deployed.

TABLE 5.4. QUIZ GENERATION EVALUATION RESULTS

Criterion	**Result**
Total generation attempts	100
Successful generation	100
Valid sentence	93/100
Grammatically correct	95/100
Meaningful Context	88/100
Unique idioms covered	22/27
Usable	88/100
Challenging	2.14/3

From the list of 119 idioms, only 27 idioms contained useful examples. This result is expected because most of the idioms in the dataset `idiom2.json` are unique and mainly focus on differentiation and are not yet available in existing example datasets. Of the 100 generated sentences, the coverage was lower, with only 22 unique idioms covered through repeated and different quiz questions, which is approximately 81.5%. Lastly, given the average topic match for three distractors per question, 71.3% matched the correct answer's topic for the specific idiom. Therefore, the quiz questions are considered challenging enough for the learners.

Sentence-Transformers

The `paraphrase-multilingual-MiniLM-L12-v2` model was evaluated based on its ability to perform semantic similarity detection. The model compares 119 idiom entries from the JSON file with the user input text instead of relying solely on exact idiom matching. As shown in the table below, most sentences correctly produce the equivalent idiom suggestion when the similarity score exceeds a predefined threshold.

An evaluation test of the Idioms in Sentences mode has been constructed to measure the accuracy, functionality, and performance of this specific mode in the application. There are two functions in the code that need to be taken into account: the idiom detection pipeline `detect_idiom` with lemma-aware matching and `detect_idioms_ai` as a fallback function when no literal match is found. Thus, a 100-sentence labeled test set[10] encompassing five categories was created:

[10]Available at https://github.com/kawaa099-dot/english-spanish-idioms-app/blob/main/evaluations/idiom_detection_test_set.csv.xlsx

TABLE 5.5. IDIOM SUGGESTION WITH MINILM

Input Sentence	**Detected Idiom**
Good luck in your upcoming tournament. Do your best!	No idioms detected
Good luck in your upcoming exam. I know you can do this, do not give up.	break a leg
Estoy muy contenta por ti.	on cloud nine
Quiero casarme con la persona que amo.	tie the knot

- Exact: the target idiom appears verbatim in its present-tense form.
- Tense: the target idiom appears in an inflected form, which is detected through lemma-aware matching.
- Semantic: the sentence captures the idiom's meaning without using its literal words, which is detected through AI similarity fallback.
- Negative: ordinary sentences with no idiom, evaluating the robustness against false positives.
- Hard Negative: sentences with literal idiom vocabulary, testing whether the system is able to distinguish between idiomatic and literal usage.

The test uses random sentences generated by Claude Sonnet 5, which are entered into the application for evaluation. Every category is checked by comparing the expected idiom and the detected idiom to determine whether they match. Table 5.6 shows examples of how the evaluation is performed.

TABLE 5.6. EXAMPLES OF IDIOM DETECTION RESULTS

Sentence	**Expected Idiom**	**Category**	**Detected**	**Match**
He always tries to break the ice at parties.	break the ice	exact	break the ice	yes
By visiting his aunt, he killed two birds with one stone.	kill two birds with one stone	tense	kill two birds with one stone	yes
He revealed the surprise without meaning to.	let the cat out of the bag	semantic	let the cat out of the bag	yes
The old dog passed away peacefully in his sleep.	kick the bucket	semantic	none	no

The output of the test is evaluated with different metrics: true positive (TP) detects the correct idiom, false negative (FN) occurs when an incorrect idiom is detected or the correct idiom is missed, and false positive (FP) occurs when an incorrect idiom is returned or when an idiom is found in a sentence where none was expected. Therefore, incorrect idiom prediction is counted as both an FP (erroneous detection) and an FN (failure to identify the true label). True negative (TN) occurs when no idiom is correctly detected in a sentence where none was expected. From these, precision, recall, and F1-score are later computed using the following formulas [77]:

$$\text{Precision} = \frac{TP}{TP + FP} \tag{5.1}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{5.2}$$

$$\text{F1-score} = 2\frac{precision \cdot recall}{precision + recall} \tag{5.3}$$

TABLE 5.7. OVERALL DETECTION METRICS (SIMILARITY THRESHOLD = 0.3)

Metric	**Value**
Precisiom	0.733
Recall	0.710
F1-score	0.721

TABLE 5.8. METRICS PER CATEGORY

Category	**n**	**TP**	**FP**	**FN**	**TN**	**Precision**	**Recall**
exact	17	17	0	0	0	1.00	1.00
tense	17	15	0	2	0	1.00	0.88
semantic	28	12	5	16	0	0.71	0.43
negative	23	—	2	—	21	—	—
hard_negative	15	—	9	—	6	—	—

For both the negative and hard_negative categories, precision or recall is not defined considering the absence of a positive class to detect. However, they can be classified through a separate table, as shown in Table 5.9 below:

TABLE 5.9. FALSE POSITIVE RATE ON NEGATIVE CATEGORIES

Category	**FP**	**TN**	**FPR**
negative	2	21	8.70%
hard_negative	9	6	60.00%

The best detection performance comes from literal idiom instances, where the Exact category obtained perfect precision and recall (100%), and the Tense category successfully achieved perfect precision (100%) and good recall (80%). Therefore, it can be concluded that the lemma-aware matching mechanism effectively generalizes across an idiom's inflected forms. Meanwhile, semantic detection was substantially weaker, with a precision of 71% and a recall of 43%, which means that the embedding-based fallback (`detect_idioms_ai`) sometimes fails to identify idiomatic meaning when the input does not contain any of the idiom's literal wording.

The False Positive Rate (FPR) for the Negative category is 8.70%, while the Hard Negative category reaches 60% for sentences using an idiom's vocabulary in a non-idiomatic sense. For example, the sentence "She climbed the ladder to fix the roof tiles" was mistakenly flagged as an idiomatic use. This is a structural limitation rather than a threshold-tuning problem: every hard-negative sentence is intercepted and incorrectly classified by the literal-matching stage itself before the AI similarity model is ever consulted, because `detect_idioms` performs literal/lemma matching first and `detect_idioms_ai` is only called when no literal match is found. Consequently, this failure mode represents a clear architectural limitation of the two-stage detection pipeline, which is further addressed in Section 6.3, and cannot be fixed by changing the similarity threshold.

Table 5.10 as follows shows how the similarity threshold from the `detect_idioms_ai` function affects the detection performance. Initially, the similarity threshold is set to 0.5 and then reduced to a value of 0.3.

TABLE 5.10. TEST RESULTS FOR DIFFERENT SIMILARITY THRESHOLDS

Metric	**Threshold = 0.5**	**Threshold = 0.3**
Precision	0.700	0.733
Recall	0.565	0.710
F1-score	0.625	0.721
semantic Recall	0.25	0.43
negative FPR	8.70%	8.70%
hard_negative FPR	60.00%	60.00%

Overall, as threshold decreasing from 0.5 to 0.3, Precision increased from 0.700 to 0.733, Recall increased from 0.565 to 0.710, and the F1-score increases from 0.625 to 0.721, with the semantic category primarily showing a significant increase, almost doubling from 0.25 to 0.43, instead of the precision/recall trade-off usually anticipated from a more liberal threshold, with a minor corresponding improvement in precision. However, both Negative FPR and Hard Negative FPR remain unchanged across different thresholds, demonstrating that threshold adjustment has no effect on these failures since they start in the literal-matching stage of the pipeline rather than the similarity-based fallback.

5.3. User Experience Evaluation

This section discusses the usability and user response to the application. For this purpose, a survey was conducted with the participation of six respondents who are related to this field and are native Spanish speakers. The survey consists of 10 questions in total: nine Likert-scale questions and one open-ended feedback question.

Interface Performance and Responsiveness. 66.7% of users agree that the interface is clear and visually organized from the first one till the last modes. They can easily access the application via laptops, desktop computers, and even mobile phones. The Streamlit interface provides fast loading times, where interactions such as translation, quiz generation, idiom detection, and analytics updates are completed within seconds. Additionally, cached resources were employed to reduce the frequency of model loading and improve overall performance. This fulfills the main goal of using Streamlit as the primary interface, which is to provide a simple means of sharing insights interactively in order to bridge the gap between end-user interaction and data analysis [78]. However, for some people it is not clearly showing the instructions very well, such as in the mode Idiom in Sentences, so they need to figure it out by themselves how that mode works.

Easy of Use. 50% of the users agree that this application is easy to navigate and very accessible. The interface of the application was designed to be straightforward and easy to use, given the simplicity of its implementation. Users can effortlessly switch between learning modes thanks to the sidebar menu, which provides access to all features. This sidebar menu makes more efficient use of the available screen space in the main layout while also providing easy navigation [79]. Moreover, users can interact with the application without requiring technical knowledge due to the minimal complexity of the interface.

Learning Effectiveness. This application supports effective learning by providing multiple forms of reinforcement, such as translations, audio pronunciation, example sentences, favorite management, idiom writing practice, quizzes, and learning analytics. Thus, users have various opportunities to improve their idiom-learning performance, which is important for effective writing and communication in daily conversations regardless of age.

66.7% of respondents rated positively (4 or 5 on the 5-point Agree scale) that this application is helpful according to its objectives, such as understanding idioms better in both languages with helpful example sentences. However, the idiom detection feature showed a poor result, with the lowest mean rating overall (3.50/5) as illustrated in the Figure 5.3. One of the respondents also commented on how this feature is not explained clearly in the app and how some of the suggested idioms were not the expected ones. This is aligned with the previous evaluation of this specific feature, where it achieved only 0.43 recall for the semantic category (with a threshold of 0.3), as shown in Table 5.10.

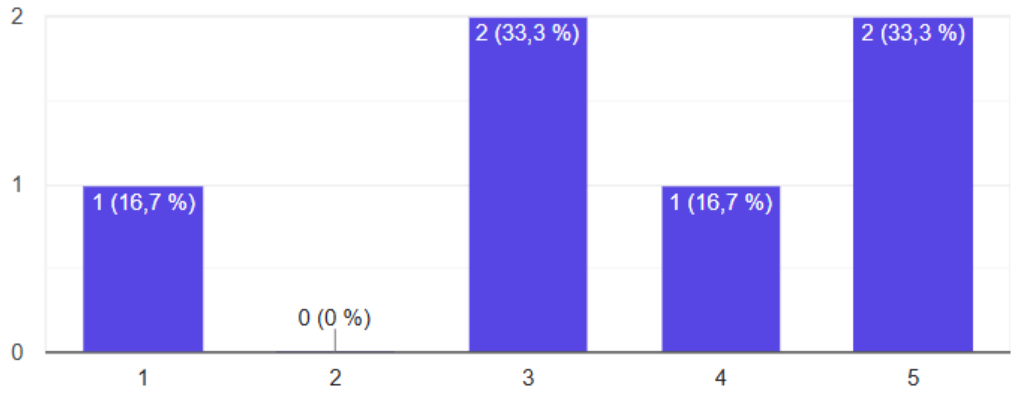


Fig. 5.3. User ratings on Idiom detection feature

AI-generated quizzes showed better results, with about 83.3% of users agreeing that this feature helps reinforce learning. However, one respondent noted that the quiz occasionally fails to generate a new question, which matches the quantitative findings in the previous section. The question of whether the application could motivate people to continue learning idioms received one disagreement (rating 1/5), one neutral response (rating 3/5), and four positive responses (two at 4/5 and two at 5/5), yielding a mean of 3.67/5, as shown in the Figure 5.4.

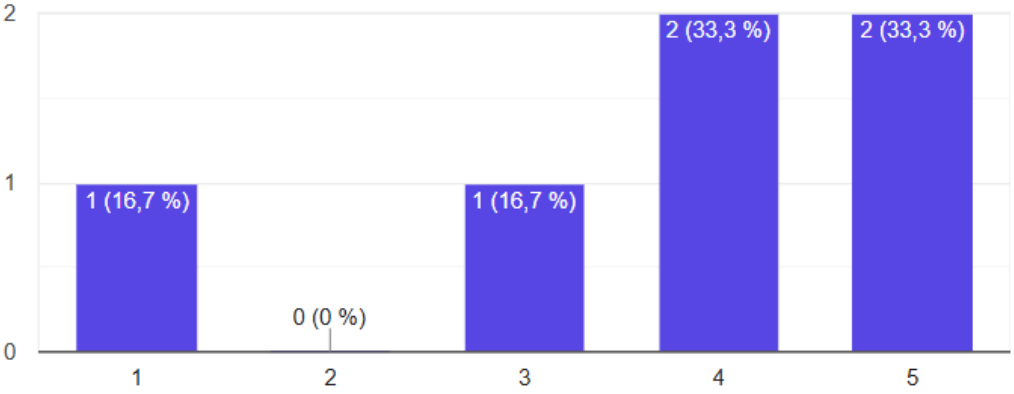


Fig. 5.4. User ratings on motivating others to learn idioms

Nevertheless, most of the respondents were satisfied with the application and would recommend it to other English learners, including the positive comment from one of the respondents in the additional comments, which was the final part of the survey.

6. CONCLUSIONS

6.1. Summary of Results

This section summarizes the results of the project based on the previously defined objectives from Section 1.2.

A JSON dataset is used as an idiom knowledge base containing English idioms together with their associated Spanish equivalent idioms or Spanish meanings, English meanings, and topic classifications (`idiom2.json`). Two example-sentence datasets were also integrated into the project as a consolidated example map, using publicly available Hugging Face datasets (IdioTS and UCSC-Admire). However, these example-sentence datasets provide examples for only 27 idioms out of the 119 idioms in `idiom2.json`, meaning that only about 22.7% of the idioms have at least one associated example sentence. This directly limits the scope of AI-generated quiz-question content, as discussed in Section 6.3.

For the literal translation component, MarianMT (`Helsinki-NLP/opus-mt-en-es`) was successfully incorporated into the application and provides valuable insight into how the natural Spanish meaning stored in the knowledge base differs from its literal translation for each idiom. Meanwhile, a dual-stage detection pipeline was created, combining lemma-aware exact and tense matching with a semantic fallback based on the `sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2` model. To evaluate the effectiveness and quality of this mechanism, a 100-sentence labeled test set was used, which showed good performance on literal and inflected idiom occurrences, as well as more constrained but non-trivial recall for semantic, paraphrased instances, as explained in Section 5.2. This indicates that the detection goal was achieved for both literal and inflected usage, with semantic detection serving as a partial rather than a complete solution. Last but not least, FLAN-T5 was used to implement an AI-assisted quiz-generation mechanism that creates original fill-in-the-blank questions by paraphrasing dataset example sentences and selecting topic-matched distractors. Paraphrased sentences proved to be more reliable than generating questions entirely from scratch. With an average distractor topic-match rate of 71.3%, an evaluation of 100 generated questions revealed that 88% were usable without manual correction, significantly surpassing the initial random-sampling baseline (0% topic-match).

For the development of the application, three items were taken into account, namely the user interface, database integration, and adaptive learning. The application was developed using the Streamlit web framework and consists of five distinct modes: Explore Idioms, Idioms in Sentences, Quiz, Favorites, and Learning Analytics. Next, SQLite was used primarily as a database to persist favorited idioms and all analytics related to each idiom across sessions, such as attempts and correct answers. Adaptive learning was in-

tegrated by prioritizing weak idioms using stored accuracy data to preferentially select idioms with sub-60% accuracy for subsequent quiz questions, thus meeting the adaptive-learning objective.

For functional testing, all primary modules were tested according to their respective functionalities and were successfully confirmed to operate as intended. In addition to functional testing, the two AI-driven components underwent quantitative evaluation. Rather than relying solely on qualitative confirmation, the literal translation component was evaluated in terms of accuracy, grammatical correctness, and context preservation, while the quiz-generation and idiom-detection mechanisms provided quantified results in terms of accuracy, precision, recall, and F1-score. In addition, a user evaluation form was distributed to several participants to assess the application, provide learning-related evaluations, and offer useful direct feedback.

6.2. Contributions

The implementation of this project has several significant contributions to AI-assisted idiom learning for English-Spanish language learners, covering both the learning tool's design and the underlying technical mechanisms.

AI-assisted idiom learning. The application integrates several existing AI technologies, which are MarianMT for literal translation, FLAN-T5 for sentence generation, and a multilingual MiniLM sentence-transformer for semantic similarity. These technologies are successfully incorporated into a single pipeline that encourages learners in idiom acquisition instead of requiring them to memorize words from static lists.

Semantic idiom recognition beyond exact matching. The common approach to idiom detection based on user text sentences relies solely on literal string matching. However, in this application, lemma-aware matching is also combined in the idiom detection pipeline in order to correctly identify idioms across inflected verb forms. For example, *"broke the ice"* is matched with *"break the ice"*, with an embedding-based semantic fallback in cases where none of the idiom's literal wording is present in the input text. This embedding-based semantic fallback can thus still recognize the idiom's meaning. The previous Section X already demonstrated this approach, which obtained near-perfect accuracy on literal and inflected-form idiom occurrences, with 1.00 Precision for both forms, 1.00 Recall for literal forms, and 0.88 for inflected forms. In addition, it expands the detection capacity to paraphrased usage that strict literal matching is completely unable to capture, albeit with reduced reliability (0.71/0.43).

Adaptive, topic-aware quiz generation. Instead of using predetermined example questions, the quiz mode uses AI paraphrasing to create new practice sentences. It also analyzes each idiom's performance to prioritize the idioms that the learner finds difficult and chooses distractors from idioms that share the same semantic theme as the correct response. This mechanism is shown in more detail in Section 5.2, focusing on substantially

increasing the number of distractors that require genuine comprehension to eliminate, compared with random distractor sampling.

A reproducible evaluation methodology for idiom-learning tools. Beyond the application itself, this project contributes a concrete, quantified evaluation methodology for AI-generated language-learning mechanisms that is able to cover both generation quality and detection accuracy. Metrics used for testing sentence generation quality include quiz sentence validity, grammaticality, and distractor difficulty. Meanwhile, detection accuracy uses known metrics such as precision, recall, and F1-score across literal, inflected, semantic, and plausible test categories. Therefore, this methodology could be replicated or extended for evaluating similar idiom or various collocation-learning systems.

Applicability. The dual detection mechanism and adaptive quiz design make it suitable for self-directed learners at intermediate to advanced English and Spanish proficiency levels, particularly those who already have enough vocabulary to benefit from contextual and semantic exposure to colloquial expressions rather than rote memorization. Additionally, the English-Spanish bilingual framing offers literal translation contrasts that highlight where idiomatic meaning extends beyond a literal word-for-word reading.

The current implementation is optimal for supplementary, self-paced practice rather than use in classroom settings. The present limitations in idiom knowledge-base coverage found in the evaluation (Section 6.3), which restrict AI-generated quiz content to the 27 idioms with available sample sentences, and the dependence on a local SQLite instance for progress tracking (Section 3.4.2) are the reasons behind this. Broader classroom or curriculum-integrated implementation would require expanding the idiom dataset to provide greater content coverage, as discussed in Future Work.

To promote replicability and additional investigation, the entire framework has been made accessible as open-source software and is available at: https://github.com/kawaa099-dot/english-spanish-idioms-app

6.3. Limitations

The project is also subject to several limitations. These include the limited idiom dataset, computational constraints that affect AI models, dependence on internet resources, limitations in semantic idiom detection, and slow AI sentence generation. Furthermore, the project focuses only on English–Spanish idiom learning, which limits its potential target audience compared with general language-learning applications. The application could also be further developed as a more robust mobile application to improve accessibility for a larger number of users.

There are several limitations that can be further discussed with respect to the development of this application, such as limited idiom dataset, limited language pair, internet resource dependency, computational constraints, MiniLM limitations, and slow AI sentence generation.

Limited Idiom Dataset. The idiom dataset integrated into this project contains only 119 idioms, which is insufficient considering the large number of idioms used worldwide. Although adequate for demonstrating the proposed learning approach, a larger dataset could enhance content diversity and provide a more comprehensive learning experience. Furthermore, out of the 119 idioms, only 27 have valid example sentences from the two datasets integrated into the application. This limitation reduces the coverage of AI-generated quiz content and example-based learning activities.

Limited Language Pair. Current implementation focuses only on English and Spanish language learning. Future work could extend the system to support additional language pairs, allowing learners with different target languages to benefit from the application and broadening its potential user base.

Internet Resource Dependency. Some components of the application depend on an internet connection to function correctly. For example, during setup and deployment, internet connectivity is required for model downloads and dataset retrieval from Hugging Face. This process may take a considerable amount of time, depending on the user's internet connection.

Computational Constraints. The project is deployed using Hugging Face Spaces and integrates Streamlit together with several AI models. As a result, it is subject to limited computational resources and CPU availability. To ensure a smooth user experience, only lightweight models were selected to maintain compatibility with the Streamlit deployment environment. Otherwise, larger language models could be employed to provide deeper semantic understanding and higher-quality text generation, thereby improving the overall effectiveness of the application.

MiniLM. The MiniLM model demonstrates good performance in the application; however, certain limitations remain. It is prone to errors when a sentence is mistakenly associated with a different idiom due to similarities between individual words, even though the overall sentence meaning is different. Moreover, if users enter long, complex, or highly technical sentences, the model may fail to detect the appropriate idiom correctly. For example, sentences containing medical terminology, legal language, or engineering specifications may reduce detection accuracy [80]. Furthermore, the suggested idiom may differ from the intended idiom while still conveying a similar meaning. When an idiom's vocabulary is employed in a non-idiomatic context, the literal and lemma-based matching step may fail to identify it. This cannot be fixed by changing the similarity threshold because it happens prior to the use of the AI-based similarity model. The literal-matching stage should be improved in the future to more clearly identify idiomatic and non-idiomatic uses.

Slow AI Sentence Generation. In the Quiz mode, the FLAN-T5-base model may take a longer time to generate new sentences due to particularly under limited computational resources . Thus, on many occasions, the system falls back to the example sentence dataset to prevent quiz interruptions while ensuring learning effectiveness. A red warning,

"Could not generate a new question. Try again.", may also appear when a new question cannot be generated, indicating that the user should try again.

6.4. Future Work

While the work demonstrates the significant potential of artificial intelligence and deep learning models for idiom learning, further research and improvements are necessary to address limitations related to language support, speech recognition, larger idiom datasets, user accounts, computational resources, and personalized recommendations.

Larger and More Diverse Idiom Datasets. Future work could expand the idiom dataset to include a more diverse and larger collection of idiomatic expressions. Hence, it would enable users to come across a wider range of idioms in a variety of contexts and themes.

Variety of Example Sentences. Future work could provide a wider range of example sentences for each idiom by integrating additional datasets or by using AI models to generate contextualized examples automatically. This would expose learners to idioms in more diverse contexts and improve learning effectiveness.

User Accounts. User accounts could be created to allow users to save their learning progress and specific preferences. Moreover, the application could be accessed across different devices without requiring users to start everything from scratch, which would promote a continuous learning experience.

Automatic Speech Recognition (ASR). ASR is a functionality that can convert spoken language into text [81]. It is a useful alternative for users who are unable to type or prefer to interact with the application through speech. Additionally, this could facilitate speaking-based learning activities and increase the application's accessibility.

Computational Resources. More effective models, model quantization, caching techniques, or hardware with higher processing power could all be investigated in future versions. This could enhance response times and offer a more seamless user experience, particularly when several AI-based features are utilized concurrently.

Personalized Recommendations. Future implementations should include broader personalized recommendations based on the learner's performance to strengthen long-term retention of idiomatic expressions. For example, a streak mechanism could encourage users to practice regularly, supporting spaced-repetition learning and enhancing their memorization of difficult idioms.

Additional Languages. The application could also incorporate more language pairs in the future, in addition to the English-Spanish language pair. This would increase the application's potential user base and enable learners with various linguistic backgrounds to benefit from it.

Offline Application. Future work could also offer an offline option, where users could

download some learning mode features to practice and learn in environments with limited internet access. It could also be available as an accessibility setting, giving more opportunities to people in areas with limited or unreliable internet connectivity.

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A. COMPLETE IDIOM DATASET

Table A.1 presents the complete set of idioms used in the application, including their Spanish meaning, English meaning, and the corresponding theme for each idiom, stored in `idiom2.json`.

This file can also be accessed and downloaded from the following link: https://github.com/kawaa099-dot/english-spanish-idioms-app/blob/main/idiom2.json

TABLE A.1. COMPLETE IDIOM DATASET (`IDIOM2.JSON`)

Idiom	**Meaning (ES)**	**Meaning (EN)**	**Topic**
break the ice	hacer que la gente se siente cómoda	To do or say something to relieve tension or awkwardness, especially when meeting someone new.	Communication
hit the sack	irse a la cama	To go to bed or go to sleep.	Daily Life
piece of cake	pan comido	Something very easy to do.	Success & Failure
once in a blue moon	de uvas a peras/muy de vez en cuando	Something that happens very rarely.	Time
actions speak louder than words	las acciones hablan más que las palabras	What people actually do matters more than what they say they will do.	Communication
a blessing in disguise	bendición disfrazada/no hay mal que por bien no venga	Something that initially seems unfortunate but later brings positive results.	Success & Failure
bite the bullet	aceptar algo díficil/armarse de valor	To face a difficult or unpleasant situation with courage.	Challenges
cost an arm and a leg	costar un ojo de la cara	To be very expensive.	Money
spill the beans	contar un secreto	To reveal a secret, often accidentally.	Communication
the ball is in your court	la pelota está en tu tejado/depende de ti	It's up to you to make the next decision or move.	Responsibility
under the weather	estar indispuesto/enfermo	Feeling ill or unwell.	Health
barking up the wrong tree	equivocarse de persona o asunto	Pursuing a mistaken course of action or blaming the wrong person.	Communication
kill two birds with one stone	conseguir dos objetivos con una sola acción	To achieve two goals with a single action.	Efficiency
let the cat out of the bag	revelar un secreto	To reveal a secret, usually without meaning to.	Communication

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TABLE A.1 – continued from previous page

Idiom	Meaning (ES)	Meaning (EN)	Topic
Hit the nail on the head	Dar en el clavo	To describe exactly what is causing a situation or problem; to be exactly right.	Communication
Bite off more than you can chew	Querer abarcar más de lo que se puede	To take on more responsibility than one can handle.	Challenges
Burn the midnight oil	Quemarse las pestañas / Trabajar hasta tarde	To work or study late into the night.	Work
Cut corners	Hacer algo a medias / Tomar atajos	To do something in the easiest or cheapest way, often sacrificing quality.	Work
Cry over spilled milk	Llorar sobre la leche derramada	To be upset about something that has already happened and cannot be changed.	Emotions
judge a book by its cover	No juzgues un libro por su portada	To evaluate someone or something based only on appearance.	Judgment
Back to the drawing board	Volver a empezar / Empezar de cero	To start a task or plan over again because the previous attempt failed.	Success & Failure
Hit the road	Ponerse en camino / Irse	To leave or start a journey.	Travel
Through thick and thin	En las buenas y en las malas	Under any circumstances, whether good or bad; loyal no matter what.	Relationships
The early bird catches the worm	Al que madruga, Dios le ayuda	People who act early or arrive first have the best chance of success.	Success & Failure
No pain, no gain	Sin esfuerzo no hay recompensa	You must work hard or suffer to achieve something valuable.	Success & Failure
Throw in the towel	Tirar la toalla / Rendirse	To give up or admit defeat.	Success & Failure
A dime a dozen	Algo muy común / Abundante	Something very common and of little special value.	Daily Life
Keep something under wraps	Mantener algo en secreto	To keep something secret or hidden.	Communication
Skeleton in the closet	Un secreto oculto / Algo vergonzoso del pasado	A hidden and embarrassing secret from someone's past.	Secrets
Straight from the horse's mouth	De primera mano / De boca del propio interesado	Information obtained directly from the original or most reliable source.	Communication

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TABLE A.1 – continued from previous page

Idiom	Meaning (ES)	Meaning (EN)	Topic
Pull your socks up	Ponerse las pilas / Esforzarse más	To make an effort to improve one's behavior or performance.	Motivation
Blood, sweat, and tears	Sangre, sudor y lágrimas / Mucho esfuerzo	Extremely hard work and effort.	Work
Break a leg	¡Mucha mierda! / ¡Suerte!	An expression used to wish someone good luck, especially before a performance.	Luck
Touch wood / Knock on wood	Tocar madera / Para atraer buena suerte	Said to avoid bad luck after mentioning something positive or hopeful.	Luck
Down on your luck	En mala racha / Con mala suerte	Experiencing a period of bad fortune, especially financial hardship.	Luck
Look before you leap	Piensa antes de actuar	Consider the consequences carefully before acting.	Decision Making
Better late than never	Más vale tarde que nunca	It's better for something to happen or be done late than not at all.	Time
Time flies	El tiempo vuela	Time passes very quickly.	Time
In the nick of time	Justo a tiempo / En el último momento	Just in time; at the last possible moment.	Time
Around the clock	Las 24 horas / Sin parar	Continuously, all day and night, without stopping.	Time
Beat the clock	Llegar antes de tiempo / Ganarle al reloj	To finish something before a deadline or time runs out.	Time
Call it a day	Dar por terminado el día / Dar por concluido	To decide to stop working on something for the day.	Work
Out of the frying pan into the fire	Salir de Guatemala y meterse en Guatepeor	To go from a bad situation to an even worse one.	Problems
Between a rock and a hard place	Entre la espada y la pared	Facing a difficult situation with only unpleasant choices.	Problems
Hit it off	Llevarse muy bien desde el principio	To immediately get along well with someone.	Relationships
Get on someone's nerves	Ponerse en los nervios / Molestar a alguien	To annoy or irritate someone.	Emotions
Bury the hatchet	Hacer las paces / Enterrar el hacha de guerra	To make peace and end a conflict or disagreement.	Relationships
Money doesn't grow on trees	El dinero no crece en los árboles	Money is not easily obtained, so it should not be wasted.	Money
Pay through the nose	Pagar un dineral	To pay an excessively high price for something.	Money

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TABLE A.1 – continued from previous page

Idiom	Meaning (ES)	Meaning (EN)	Topic
Penny-wise and pound-foolish	Ahorrar en lo pequeño y gastar en lo grande / Tacaño e imprudente	Careful with small amounts of money but wasteful with large amounts.	Money
Blow off steam	Desahogarse / Sacar la rabia	To release pent-up energy, tension, or anger.	Emotions
On cloud nine	En el séptimo cielo / Muy feliz	Extremely happy.	Emotions
Go ballistic	Perder los estribos / Enfurecerse	To become extremely angry very suddenly.	Emotions
Wear your heart on your sleeve	Mostrar tus sentimientos abiertamente	To openly show one's emotions.	Emotions
Step up to the plate	Dar un paso al frente / Asumir la responsabilidad	To take responsibility or action when needed.	Responsibility
Play it by ear	Improvisar / Actuar según la situación	To improvise or decide how to act as a situation develops, without a set plan.	Decision Making
Think outside the box	Pensar de manera creativa / Romper esquemas	To think creatively, beyond conventional ideas.	Creativity
Climb the ladder	Ascender en la carrera profesional	To advance in one's career or social status.	Work
Call the shots	Tomar las decisiones / Llevar la batuta	To be in charge and make the decisions.	Leadership
Hit the ground running	Empezar con fuerza / Ponerse manos a la obra	To start a new activity or job with energy and immediate success.	Work
Back on your feet	Recuperarse / Volver a estar en pie	To recover, especially after illness or financial trouble.	Health
Out of sorts	Sentirse mal / De mal humor	Feeling slightly unwell or in a bad mood.	Health
Fit as a fiddle	En plena forma / Muy saludable	In excellent physical health.	Health
Burn out	Agotarse / Quemarse (por exceso de trabajo o estrés)	To become exhausted, physically or mentally, from overwork or stress.	Health
Curiosity killed the cat	La curiosidad mató al gato	Being too curious about something can get you into trouble.	Animals
Let sleeping dogs lie	No despertar al perro que duerme / No remover problemas antiguos	To avoid interfering with a situation that could cause problems if disturbed.	Problems
The lion's share	La mayor parte / La porción más grande	The largest part of something.	Animals

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TABLE A.1 – continued from previous page

Idiom	Meaning (ES)	Meaning (EN)	Topic
Birds of a feather flock together	Dios los cría y ellos se juntan / Dime con quién andas...	People with similar interests or characters tend to associate with each other.	Relationships
Hold your horses	Calma / Espera un momento	Wait, be patient, or slow down.	Animals
Bring home the bacon	Ganar el sustento / Mantener a la familia	To earn money to support one's family.	Work
Apple of one's eye	El ojito derecho / Muy querido	Someone who is cherished above all others.	Relationships
Every cloud has a silver lining	No hay mal que por bien no venga	Every difficult situation has some positive aspect.	Optimism
Rain on someone's parade	Arruinarle la fiesta a alguien / Echar a perder algo	To spoil someone's plans or happiness.	Emotions
Chase rainbows	Perseguir ilusiones / Buscar algo imposible	To pursue an unrealistic or unattainable goal.	Dreams
Come rain or shine	Pase lo que pase / Con lluvia o sol	Regardless of circumstances or difficulties; no matter what happens.	Commitment
Go the extra mile	Hacer un esfuerzo adicional / Ir más allá	To make an extra effort beyond what is expected.	Motivation
Off the beaten track	Fuera de lo común / En un lugar remoto	In a remote or unusual place, away from popular routes.	Travel
Take the plunge	Dar el salto / Atreverse	To decide to do something risky or important after hesitating.	Decision Making
Make a long story short	Para no hacer larga la historia / En resumen	To summarize; to give the essential facts without unnecessary detail.	Communication
Take with a grain of salt	Tomar con pinzas	To view something with skepticism; not to take it entirely at face value.	Communication
Throw caution to the wind	Echar la cautela al viento / Arriesgarse	To act recklessly, ignoring risk.	Risk
Cross that bridge when you come to it	No te preocupes antes de tiempo	To deal with a problem only when it actually happens, not before.	Problems
In hot water	En apuros / Metido en problemas	In trouble or a difficult situation.	Problems
Playing with fire	Jugar con fuego / Tomar riesgos innecesarios	Taking a dangerous risk.	Risk

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TABLE A.1 – continued from previous page

Idiom	**Meaning (ES)**	**Meaning (EN)**	**Topic**
Bite the hand that feeds you	Morder la mano que te da de comer	To harm or betray someone who has helped or supported you.	Relationships
Jack of all trades, master of none	Aprendiz de todo, maestro de nada	Someone who can do many things reasonably well but is not an expert in any of them.	Work
Keep your eyes peeled	Mantener los ojos bien abiertos / Estar atento	To watch carefully or stay alert for something.	Awareness
Jump on the bandwagon	Subirse al carro / Seguir la tendencia	To join others in doing something popular or fashionable.	Society
Let bygones be bygones	Lo pasado, pasado / Dejar atrás rencores	To forget past conflicts or offenses and move forward.	Relationships
Walk on very thin ice	En terreno peligroso / Al borde del problema	To be in a risky or precarious situation.	Risk
Pull someone's leg	Tomar el pelo a alguien	To tease or joke with someone, often by telling them something untrue.	Humor
Up in the air	En el aire / Sin decidir	Undecided or uncertain.	Decision Making
Wash your hands of something	Lavar las manos / Evitar responsabilidades	To refuse to take further responsibility for something.	Responsibility
The ship has sailed	se te pasó el tren / ya es demasiado tarde	An opportunity has passed and is no longer available.	Time
Walking on eggshells	andar con pies de plomo / actuar con mucho cuidado	Being very careful not to upset or offend someone.	Relationships
Working your fingers to the bone	trabajar como un burro / trabajar muy duro sin descanso	Working extremely hard, often to the point of exhaustion.	Work
Speak of the Devil	hablando del rey de Roma	Said when a person being discussed suddenly appears or is mentioned.	Communication
Green thumb	tener buena mano para las plantas	A natural talent for gardening and growing plants.	Hobbies
On the fence	estar entre dos aguas / estar indeciso	Undecided about something.	Decision Making
Ring a bell	sonar familiar / resultar conocido	To sound familiar; to remind someone of something.	Memory
Black eye	ojo morado	A bruise around the eye, typically caused by a blow.	Health

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TABLE A.1 – continued from previous page

Idiom	Meaning (ES)	Meaning (EN)	Topic
Turning the tables	darle la vuelta a la tortilla / dar la vuelta a la situación	To reverse a situation completely, especially one's own disadvantage into an advantage.	Success & Failure
Wearing many hats	desempeñar muchos papeles / hacer de todo	Performing several different roles or jobs at once.	Work
Old school	tradicional / de la vieja escuela	Traditional or conventional in style or values.	Society
Kick the bucket	estirar la pata / morir	To die.	Life & Death
Hot potato	tema espinoso o delicado	A controversial or sensitive issue that is difficult to deal with.	Problems
Showing one's true colors	mostrar su verdadera cara	Revealing one's real character or intentions, often after pretending otherwise.	Personality
Turn a blind eye	hacer la vista gorda	To deliberately ignore something wrong or improper.	Responsibility
Tie the knot	casarse	To get married.	Relationships
Knock someone's socks off	dejar a alguien con la boca abierta / impresionar muchísimo a alguien	To impress or amaze someone greatly.	Emotions
A stone's throw away	a tiro de piedra / muy cerca	Very close in distance.	Travel
Sit tight	esperar pacientemente	To wait patiently and not take action.	Patience
Drive someone nuts	volver loco a alguien / sacar de quicio	To annoy or irritate someone intensely.	Emotions
Back to square one	volver a la casilla de salida / volver al punto de partida	Back to the starting point, as if no progress had been made.	Success & Failure
Cut the mustard	estar a la altura / cumplir las expectativas	To meet expectations or perform to the required standard.	Success & Failure
Cold turkey	dejar algo de golpe y sin ayuda	To stop doing something (often an addictive habit) suddenly and completely, without gradual reduction.	Health
Be a night owl	ser noctámbulo	To be someone who is naturally more active or alert late at night.	Daily Life
Be over the moon	estar en las nubes de felicidad / estar muy feliz / encantado	To be extremely happy or delighted.	Emotions

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TABLE A.1 – continued from previous page

Idiom	**Meaning (ES)**	**Meaning (EN)**	**Topic**
Beat around the bush	andar por las ramas / no ir al grano	To avoid talking about something directly; to speak evasively.	Communication

B. PLANNING

B.1. Task Description

- Task 1 – Project Definition: Choosing the project topic, defining the objectives, and conducting research on idiom learning and available AI technologies.
  - Task 1.1 – Problem identification: Planning a problem and how to solve it.
  - Task 1.2 – Literature Review: Review existing language-learning applications, idiom-learning resources, and NLP technologies.
  - Task 1.3 – Getting familiar with AI models, Large Language Model and Hugging Face
- Task 2 – Data Collection: Collecting data from different sources, processing the data, and integrating it into the project.
  - Task 2.1 – Dataset Acquisition: Collect idiom datasets from OpenAI ChatGPT and example sentence datasets from Hugging Face.
  - Task 2.2 – Exploring the data: Understand the type of data provided, such as idiom meanings, translations, associated topics, and example sentences.
  - Task 2.3 – Cleaning the data: Perform idiom normalization, remove duplicates, and standardize data formats.
  - Task 2.4 – Data Integration: Idioms and their natural translations are gathered into a JSON file, while the example datasets are combined into a unified JSON structure.
- Task 3 – Application Development: Planning the design of the application, considering both front-end and back-end development, and the implementation of the database.
  - Task 3.1 – Front-end Development: Design and implementation of the user interface using Streamlit, with integration of Hugging Face models and datasets.
  - Task 3.2 – Back-end Development: Implementation of Explore Idioms, Translations, Favorites, Quiz Generation, and Learning Analytics functionalities.
  - Task 3.3 – Database development: Implement an SQLite database to store favorites, quiz performance, and learning progress.
- Task 4 – AI integration: Integration of several AI models into the application based on their respective functionalities.

  - Task 4.1 – MarianMT Integration: Implement literal English–Spanish idiom translation.
  - Task 4.2 – FLAN-T5 Integration: Implement AI-assisted quiz generation.
  - Task 4.3 – Sentence Transformer Integration: Deploy semantic similarity matching for idiom recognition.

- Task 5 – Testing and Evaluation: Test and evaluate the functionality of each mode of the application, while also taking user experience into account.
  - Task 5.1 – Functional Testing: Verify the functionality of the application features.
  - Task 5.2 – Model Evaluation: Assess the quality of the translations, quizzes, and idiom detection.
  - Task 5.3 – User Experience Evaluation: Evaluate the usefulness of the application with respect to its educational objectives and overall usability, particularly as an idiom-learning application.

- Task 6 – Preparation of the Final Report: Compile the work into a final report and prepare for the project defense.
  - Task 6.1 – Planning and Structure of the Report: Design the structure of the final report and plan its content.
  - Task 6.2 – Draft of the report: Write the full initial draft of the report document.
  - Task 6.3 – Review and Revision of the Report: Edit, polish, and refine the report based on the advisor's feedback to produce the final version.
  - Task 6.4 – Defense Preparation: Create a presentation that summarizes the project and provides sufficient evidence to support the work. This task also includes practicing presentation delivery and preparing for potential questions.

B.2. Task Sequencing

The sequencing of tasks represents the time allocated to each task described in Part B.1. Some tasks are interrelated, as indicated by the Dependencies column as shown in the table below.

TABLE B.1. TASK DURATIONS AND DEPENDENCIES

Task	**Duration**	**Dependencies**
Task 1 - Project Definition		
Task 1.1 - Problem identification	2 Weeks	
Task 1.2 - Literature Review	2 Weeks	1.1
Task 1.3 - Getting familiar with AI models, Large Language Model and Hugging Face	4 Weeks	1.2
Task 2 - Data Collection		
Task 2.1 - Dataset Acquisition	3 Weeks	1.2,1.3
Task 2.2 - Exploring the data	2 Weeks	2.1
Task 2.3 - Cleaning the data	1 Week	2.2
Task 2.4 - Data Integration	2 Weeks	2.3
Task 3 - Application Development		
Task 3.1 - Front-end Development	3 Weeks	2.4
Task 3.2 - Back-end Development	5 Weeks	2.4
Task 3.3 – Database development	2 Weeks	3.2
Task 4 - AI integration		
Task 4.1 - MarianMT Integration	1 Week	3.2
Task 4.2 - FLAN-T5 Integration	2 Weeks	3.2
Task 4.3 - Sentence Transformer Integration	2 Weeks	3.2
Task 5 - Testing and Evaluation		
Task 5.1 - Functional Testing	3 Weeks	3.1, 3.2, 3.3, 4.1, 4.2, 4.3
Task 5.2 - Model Evaluation	2 Weeks	4.1,4.2,4.3
Task 5.3 - User Experience Evaluation	2 Weeks	5.1,5.2
Task 6 - Preparation of the Final Report		
Task 6.1 - Planning and Structure of the Report	2 Weeks	
Task 6.2 - Draft of the report	12 Weeks	6.1
Task 6.3 - Review and Revision of the Report	2 Weeks	6.1,6.2
Task 6.4 - Defense Preparation	1 Week	6.1,6.2,6.3

B.3. Gantt Chart

The Gantt chart provides a graphical overview of the project timeline, showing the duration and scheduling of each task throughout the development process. The project spans a total duration of 49 weeks, from 6 October 2025 to 14 September 2026. All tasks under Task 4 (AI Integration) are dependent on Task 3.2 (Back-End Development). Therefore, upon completion of the back-end architecture, the three AI models can be integrated in parallel.

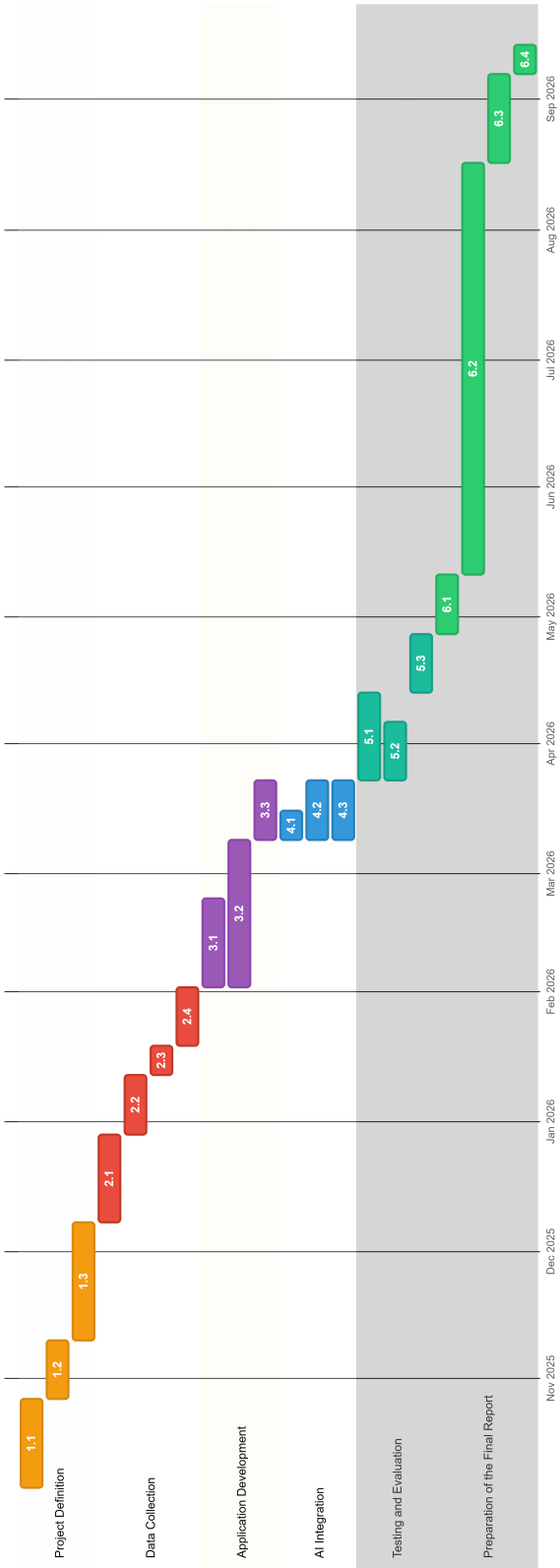


Fig. B.1. Gantt chart

C. BUDGET

C.1. Human Resources

This project has been carried out by a university student pursuing a Bachelor's Degree in Telecommunication Technologies Engineering at Universidad Carlos III, under the supervision of her tutor, Pedro Manuel Moreno Marcos. The costs associated with the human resources involved were estimated based on the hours contributed by both parties.

The project spanned approximately 55 weeks starting in November 2025 till September 2026. During this period, a total of 18 meetings were conducted, with an average duration of one hour per meeting, resulting in approximately 18 hours of meetings. Additionally, considering the extra time spent by the advisor reviewing drafts and providing feedback via email, approximately 48 hours of supervision were contributed to the project.

Assuming that the hourly rate for an experienced engineer in Madrid is approximately 24€ per hour, according to Glassdoor, and considering a 40-hour work week, this corresponds to a monthly salary of approximately 3,840€. Therefore, the advisor's estimated cost, considering a total of 48 hours of work, is 1,152€.

In contrast, the hourly rate for the student was estimated at approximately 16€ per hour. The student developed the project over a total of 55 weeks, dedicating approximately 8 hours per week to its completion, resulting in total of 440 hours.

In conclusion, the student's contribution in terms of human resources is estimated at 7,040€.

C.2. Hardware and Software Costs

The project was developed using an ASUS VivoBook 14 equipped with an 11th Gen Intel® Core™ i5 processor and 8 GB DDR4 RAM. The laptop was purchased for RM 2,999, which is equivalent to approximately 635.73€ based on the current exchange rate. Taking the project duration into account, the depreciation cost of the laptop can be estimated as shown below.

$$\text{Amortization} = \left(\frac{\text{cost} \times \text{time of use}}{\text{lifetime}}\right) = \left(\frac{635.73 \times 1.06}{5}\right) = 134.77\ €$$

The amortization period is assumed to be 5 years, while the duration of use for this project was 55 weeks, equivalent to approximately 1.06 years. Therefore, the amortization cost attributable to the project is 134.77€.

Apart from the hardware, several open-source software tools were used throughout

the project at no additional cost:

- **Google Colab**: A free cloud-based environment for Python programming, where the project was initially developed using Jupyter Notebooks.
- **VS Code**: A free source-code editor developed by Microsoft, used for coding, debugging, and testing.
- **Hugging Face Spaces**: A free platform used to deploy the final application.
- **Overleaf**: A free online LaTeX editor used for report writing and collaboration.
- **Streamlit**: Graphical user interface development.
- **SQLite**: Local database management.
- **MarianMT**: English–Spanish literal translation.
- **FLAN-T5**: AI-assisted quiz generation.
- **Multilingual MiniLM**: Semantic idiom detection.
- **gTTS**: Text-to-speech functionality.
- **Hugging Face Datasets**: Example sentences and idiom datasets.

All models, libraries, and software tools used in this project were free and publicly available; therefore, they do not contribute to the software costs. However, other consumable resources such as internet connectivity, electricity, and miscellaneous supplies should also be considered. These expenses are estimated at approximately 10% of the total project cost.

C.3. Total Cost

Table C.1 summarizes the total cost of the project, obtained by summing all human resource costs, material costs, including hardware and software, and miscellaneous expenses. The total project cost is calculated to be 9159.45€.

TABLE C.1. TOTAL COST

Type	Description	Cost
Human Resources	Tutor	1152€
Human Resources	Student	7040€
Material Costs	Hardware	134.77€
Material Costs	Software	0
Cost of Consumables	Other (10%)	832.68€
Total cost		**9159.45€**

D. REGULATORY FRAMEWORK

The development and implementation of this project have been guided by multiple regulations and legal requirements. Legal and ethical frameworks are associated with the use of open-source software, pre-trained artificial intelligence models, and datasets, all of which support the primary objective of this project. These frameworks help ensure compliance with copyright regulations, licensing requirements, and accepted standards for data usage.

Through the Hugging Face platform, several datasets and pre-trained Natural Language Processing (NLP) models are publicly available. The example sentence datasets used in this project were obtained from two different sources, each distributed under a different license. The fdelucaf/IdioTS dataset is distributed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 (CC BY-NC-SA 4.0) license, which permits use, modification, and redistribution for non-commercial purposes, provided that appropriate attribution is given and derivative works are distributed under the same license [82]. The dataset was used solely for educational and research purposes throughout this project.

In addition, the UCSC-Admire idiom dataset was used as a complementary source to provide additional example sentences for educational purposes. The project also provides proper attribution to the original source of the dataset.

There are several open-source technologies that have been incorporated into this project, such as Python, the Hugging Face Transformers library, Streamlit, SQLite, gTTS, MarianMT, FLAN-T5, and MiniLM. These tools are provided under open-source licenses so that they can be used, modified, and redistributed for research and educational purposes. Open-source software helps future developers and researchers reduce development costs while improving transparency, reproducibility, and accessibility.

Regarding copyright considerations, every external source, figure, and technical data used in this work is duly cited in the relevant parts of the report. Any images, diagrams, or resources obtained from third-party sources are used in accordance with their respective licenses and attribution requirements.

Throughout the development of this project, the application does not collect, store, or process personally identifiable information. The system implements an SQLite database that solely stores learning-related information, including favorite idioms, quiz attempts, and learning analytics, with no transmission, collection, or distribution of sensitive personal data. Moreover, the example datasets also do not contain any personal information, therefore reducing concerns regarding the handling of sensitive information. As a result, the application presents minimal privacy risks.

Through the integration of open and free technology, the project also helps to support educational inclusion and accessibility. Deploying the application on Hugging Face

Spaces enables users to access the system through a web browser without requiring specialized hardware or paid software licenses. Thus, this application promotes an accessible and interactive language-learning experience through existing features, including bilingual explanations, audio pronunciation, AI-assisted idiom detection, and adaptive quizzes.

In summary, the project respects intellectual property rights, adheres to responsible data-use policies, meets relevant licensing requirements, and promotes accessibility through the use of open-source technologies. The use of free and openly accessible tools also enhances the reproducibility, transparency, and educational value of the proposed system.

E. SOCIO-ECONOMIC FRAMEWORK

The work carried out in this project presents several socio-economic benefits for both language learners and educators, including educational institutions, with the broader goal of supporting cross-cultural communication. The system offers AI-integrated idiom learning in a more accessible and interactive way, addressing one of the most challenging aspects of learning a foreign language.

The application itself enables the learning and practice of idiomatic expressions through multiple learning options, such as adaptive quizzes, interactive idiom exploration, and AI-assisted sentence generation. Hence, it supports learners who lack access to physical classes or human tutors. From an economic perspective, this application may provide greater accessibility to rural and underserved areas in the domain of language learning, particularly idiomatic and figurative competence, which is often excluded from formal language courses despite its importance in real-world communication. The tool reduces the financial barriers associated with private tutoring or specialized conversation sessions by offering free, on-demand practice. Therefore, it particularly benefits learners who cannot afford supplementary language training. This can contribute to improved language proficiency and communication skills, which are crucial in both academic and professional environments.

One of the main modes that has a significant impact on learners is the adaptive quiz system. The quizzes help learners improve their performance by taking their weaknesses into account and prioritizing them during practice. Therefore, they provide a valuable alternative to traditional learning materials that rely on non-adaptive methods.

Educators and educational institutions may also benefit from a tool of this kind. One of the more time-consuming aspects of language teaching is explaining and evaluating idiomatic and figurative language, as it often requires case-by-case explanations of cultural context and usage. A tool that automates the identification, explanation, and practice of idioms can provide additional learning opportunities while reducing the teaching workload. This allows educators to devote more classroom time to conversation practice, writing feedback, and other instructional activities. Meanwhile, institutions teaching English or Spanish could integrate such a tool to support student learning without a proportional increase in staffing costs.

Furthermore, this application can significantly help students improve their communication skills despite language barriers, which is important in both educational and professional environments.

From an economic perspective, the project offers a more affordable educational tool for schools, academic institutions, and independent learners. This is achieved through the use of open-source software, freely accessible AI models, and publicly available datasets.

This initiative helps provide more equal access to a level of language competence that is often difficult to acquire, regardless of a learner's socioeconomic background or geographic location, by offering a free and self-paced learning alternative.

On a broader scale, this application helps reduce communication barriers in society and promotes cross-cultural understanding and idiomatic language proficiency. For example, it can help users avoid misinterpretations of figurative language or unintended offense that may occur in daily interactions, whether in personal or professional environments. This is particularly important in an increasingly interconnected world where remote work, international collaboration, and cross-border communication are becoming more common.

The aforementioned benefits align with the United Nations SDG[11] (Sustainable Development Goal) 4, Quality Education, by providing accessible technology-based learning opportunities that support continuous learning beyond the traditional classroom environment. In addition, the project aligns with Goal 8: Decent Work and Economic Growth by enhancing learners' employability and communication skills in globalized professional contexts. Lastly, the project also contributes to Goal 10: Reduced Inequalities, as reducing financial barriers to specialized language education helps democratize access to linguistic and cultural competence that has traditionally depended on costly immersion experiences or private tutoring.

[11] https://unstats.un.org/sdgs/report/2024/

F. DECLARATION OF USE OF GENERATIVE AI

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DECLARATION OF USE OF GENERATIVE ARTIFICIAL INTELLIGENCE (AI) IN BACHELOR THESIS (TFG)

I have used Generative AI in my TFG

Check all that apply:

YES	NO

If you have ticked YES, please complete the following 3 parts of this document:

Part 1: Reflection on legal, ethical and responsible behavior

Please be aware that the use of AI carries some risks and may lead to a series of serious academic consequences: the evaluation of your TFG could be compromised if the use of AI involves the use of confidential data, copyrighted materials, or personal data, used without complying with the conditions required in each case (authorization of the interested parties, authorization of the owners, following the instructions of the University).

Question	
1. In my interactions with AI tools, I have provided **confidential data**, always with the appropriate authorization of the data subjects. Confidentiality encompasses any information that a person or organization wishes to protect for legal, commercial, privacy, or strategic reasons (such as patents or trade secrets).	
YES, I have used this data with permission	NO, I have not used confidential data
2. In my interaction with Generative AI tools, I have submitted **copyrighted materials** with the permission of those concerned.	
YES, I have used these materials with permission from the copyright holders, or without permission because they fall within one of the exceptions or limitations permitted by law: • Public domain work • Licensed work (Creative Commons licenses) • Use of excerpts for research purposes (fair use)	NO, I have not used copyrighted materials

3. In my interaction with Generative AI tools, I have submitted **personal data** with the consent of the data subjects.	
YES, I have used this data with the authorization of the interested parties and in accordance with the instructions contained in the guide approved by the University.	NO, I have not used personal data
4. My use of the Generative AI tool has **respected its terms of use**, as well as the essential ethical principles, not being maliciously oriented to obtain an inappropriate result for the work presented, that is to say, one that produces an impression or knowledge contrary to the reality of the results obtained, that supplants my own work or that could harm people.	
YES	NO

Part 2: Declaration of technical use

Use the following model statement as many times as necessary, in order to reflect all types of iteration you have had with Generative AI tools. Include one example for each type of use where indicated: *[Add an example].*

I declare that I have made use of the Generative AI system (Name of AI system/tool and version: ChatGPT, Gemini, Copilot...) **for:**

Documentation and drafting:

- *Supporting reflection in relation to the development of the work: iterative process of analysing alternatives and approaches using AI*

[Add an example] For example: I have asked for a list of alternatives to address the problem stated at the beginning of the paper.
I have asked ChatGPT some alternatives of tools that I am going to use in the project.

- *Revision or rewriting of previously drafted paragraphs*

[Add an example] For example: I have asked for paragraphs to be rewritten to shorten the conclusions while keeping the same ideas already written.
I have asked synonym of some words to be more academically suitable for the theses using ChatGPT 5.6. I also have asked Gemini 3.6 to help me to do LaTeX formatting for listing and table.

- *Search for information or answers to specific questions*

[Add an example] For example: The questions asked were aimed at [finding definitions, explanations of processes, examples of use, lists of risks of use...].

- *Bibliography search*
- *Summary of bibliography consulted*
- *Translation of texts consulted*

Develop specific content

Generative AI has been used as a support tool for the development of the specific content of the TFG, including:

- *Assistance in the development of lines of code (programming)*

[Add an example of the usual flow of use for programming] For example: I have asked for explanations about a code in Python.
I have asked for explanations about a code in Python that I found during my research while developing the project.

- *Generation of diagrams, images, audios or videos*

[indicate its use in this section, as well as in a footnote to the figure/element generated by Generative AI] For example: Source: own elaboration with the help of Midjourney.

- *Optimisation processes*

[Add an example of the usual flow of use to optimize] For example: I have asked for help to correct and explain errors in a code that was not working properly.
I have asked for help from Claude Sonnet 5 Medium to explain errors in a code that I not understand clearly.

- *Data processing: collection, analysis, cross-checking of data...*

[Add an example] For example: after collecting the data, I have asked for guidance in normalizing the data in preparation for the subsequent analysis.

- *Inspiration of ideas in the creative process*

[Add an example] For example: once I was clear about the objective of the work, I asked for ideas to cover the project from a multidisciplinary approach.
I use Gemini 3.6 to help me with Gantt chart created using mermaid.ai

- *Other uses linked to the generation of specific points of the specific development of the paper*

If you think it is necessary to include it, add it in each case:

using the *prompt* ...
(Write the request made to the AI)
having as interaction ...
(Describe what interaction you made with the AI after the prompt's response)

Example: ***I declare that*** *I have made use of the Generative AI system* ***Chat GPT X.x*** *to* ***search for information*** *using* ***the prompt: "Tell me a concrete example that illustrates the application of the urban planning proposal of the 15-minute City in Spain"****, having as interaction* ***the proposition of the concrete example of the Chamartín district that has been reflected in the dissertation****.*

Part 3: Reflection on utility

Please provide a personal assessment (free format) of the strengths and weaknesses you have identified in the use of Generative AI tools in the development of your work. Mention if it has helped you in the learning process, or in the development or drawing conclusions from your work.

Generative AI tools helped me start from a blank page, allowing me to be more productive in my work. They also provided assistance with sentence construction and coding tasks, although their outputs were not always reliable. To complete this project, I still needed to apply critical and analytical thinking, supported by broader research and access to university library resources, in order to define and develop a higher-quality piece of work.